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Greening Energy Market and Finance

**Volumetric analysis of wind power
production for Power Purchase Agreement
(PPA) purposes**

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Contents

1	Introduction	5
2	Wind power: history, functioning and predictions	7
2.1	Technical functioning of a wind turbine	7
2.2	Advantages and disadvantages of wind power generation	9
2.3	Forecasting models for wind power generation	10
3	The Data	13
3.1	Betz Law	13
3.2	Energy production by using power curves	15
3.3	Historical production data analysis	16
3.3.1	Weibull	16
3.3.2	Seasonality	17
3.3.3	Energy produced and statistical measures	19
4	Forecast	23
4.1	Monte Carlo Simulation	23
4.1.1	Monte Carlo Modeling	25
4.1.2	Results	27
4.2	Time series analysis	28
4.2.1	Backshift operator	29
4.2.2	Differentiation	29
4.2.3	Invertibility	30
4.3	Autoregressive model (AR)	31
4.4	Moving Average model (MA)	32
4.5	Mixed Autoregressive-Moving Average Models	32
4.5.1	Autoregressive Moving Average (ARMA)	32
4.5.2	Autoregressive Integrated Moving Average (ARIMA)	33
4.6	Time series modeling	34
4.6.1	Identification of potential explanatory variables	34
4.6.2	Identification and management of the seasonal component of the time series .	37
4.6.3	Data manipulation	40
4.6.4	Stationarity test	41
4.6.5	Autocorrelation Function (ACF)	42
4.6.6	Partial Autocorrelation Function (PACF)	43

4.6.7	Model identification	45
4.6.8	Model fitting and forecasting	46
4.6.9	Downscaling Process	51
4.6.10	Results	52
5	Comparison	57
6	Conclusion	63

Chapter 1

Introduction

The purpose of this thesis is to make a prospective assessment of the wind energy that will be produced so that this dimension can be included in the definition of a Power Purchase Agreement (which we will henceforth call a PPA). A PPA is a long-term contract form involving two parties, the clean energy producer and a consumer/buyer (referred to as an off-taker). This contract form has different configurations since:

- Where the buyer is a company/entity, then we will talk about what is called CPPA (Corporate Power Purchase Agreement). Otherwise, where the buyer is a power trader, we will talk about Merchant Power Purchase Agreement.
- In case the plant was built on the buyer's owned area and this plant is placed upstream of the meter itself then we will talk about on-stream PPA. Otherwise, in case the plant was built on an area owned by the supplier and the power was distributed through national power grid, we will talk about off-stream PPA.
- Finally, a distinction concerns the type of assessment. Indeed, if a physical assessment is configured the energy will actually be exchanged between the producer and the buyer and we will be discussing about a PPA on the other hand, if the assessment is virtual, thus representing a purely financial transaction, we will talk about VPPA (Virtual Power Purchase Agreement).

Since, as previously mentioned, a PPA is a long-term contract (ten to fifteen years on average), the forecasting analysis about the amount of energy that will be produced appears to be crucial to the proper definition of the contract.

Chapter 2

Wind power: history, functioning and predictions

Wind power has historically been used as early as ancient civilizations exploiting its potential for navigation. In modern times, starting in the 1980s with the construction of the first wind farm (in the United States) for commercial power generation, wind power has always been widely used for power generation.

After the success of the first wind farm such industry has experienced a strong development all over the world expanding in Europe, and especially in Denmark Germany and Spain. It will be Europe that will be the leader in the sector until the passage in China of the law inherent in renewable energy (2005). (Source: deCastroa *et al.*).

In recent years, thanks to spatial configuration, strategic investment and supportive policies, several European countries (such as Ireland, Portugal, Germany, Spain and Italy) have distinguished themselves by the strong development of the wind power sector.

It is precisely Italy that has experienced and is continuing to experience a strong phase of development in this sector, configuring itself as the fifth largest country in Europe in terms of installed capacity (to date 12.3 GW and with projections for 2030 that foresee an installed capacity of 28GW net of any divestment of old plants).

Wind power production in Italy is mostly concentrated in Apulia (with 3.1 wind turbines), Sicily (with 2.2 GW installed) and Campania (with 1.9 GW installed). Our study will focus precisely on Sicily, which, due to its configuration, geographic location, and the wide possibility of building both in-shore and off-shore plants (the latter corresponding to plants located and built on platforms in the sea) seems to be a focal point of interest for wind energy development on Italian country.

2.1 Technical functioning of a wind turbine

The electricity produced by wind turbines is the result of converting wind kinetic energy into mechanical rotational energy. When the wind speed (kinetic energy) reaches the minimum starting threshold (of about 3 m/s), the rotor, consisting of three blades, begins its movement (mechanical energy), which increases proportionally with the wind speed until it reaches a maximum threshold when a wind speed of 25 m/s is reached. The rotor is connected to the nacelle via the slow shaft, which rotates at the same speed as the blades (up to a maximum of 16 rotations per minute). The mechanical energy is

first amplified by the RPM multiplier (by about 100 times) and then transferred to the generator, inside which, through a system of magnets, the mechanical energy is transformed into electrical energy. The electrical energy is produced by the generator at low voltage (LV) and, using a transformer inside the turbine, is elevated to medium voltage (MV).

Finally, the electricity produced by the wind turbines that constitute the wind farm is transported via underground conduits to a point of connection to the national transmission grid. Through a transformer placed at the connection point, the electricity is usually converted to high voltage (HV) (this is since for the same power transported, there is greater efficiency and reduced dissipation losses). In the image below, it is possible to observe all the components of a wind turbine previously illustrated and other components not directly included in the power generation process but essential to the proper operation of a plant, such as:

- **Brake:** Functional component for slowing blade rotation in excessive wind situations. This component turns out to be essential in order to ensure that the system does not suffer damage in extreme conditions.
- This component plays a key role in controlling and adjusting the angle and speed of rotation of the blades, optimizing energy production and preventing damage to the plant itself. In addition, this instrument is also used for measuring wind speed, allowing the monitoring of the plant's performance over time.

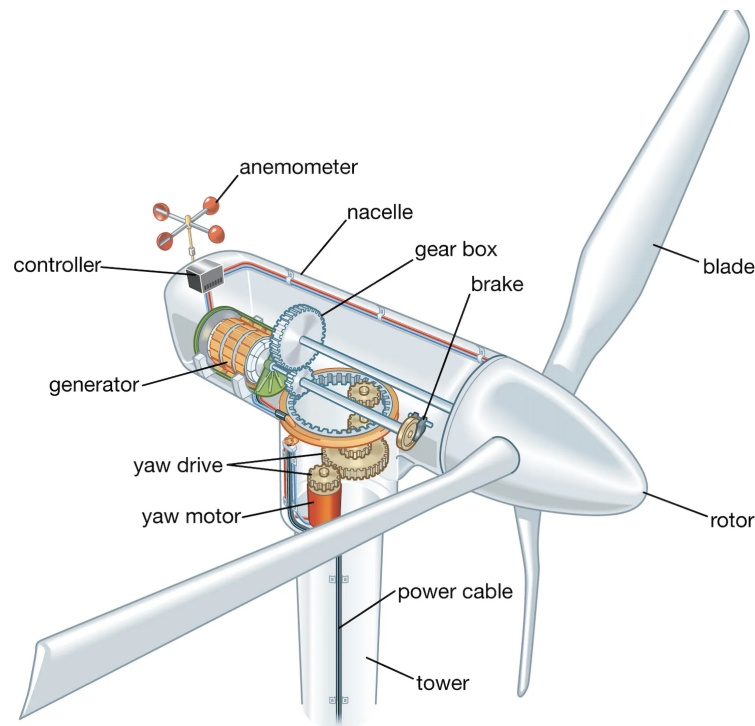


Figure 1: Wind turbine components.
(Source: Encyclopedia Britannica)

2.2 Advantages and disadvantages of wind power generation

Wind energy, as well as all energy from renewable sources, has numerous benefits and disadvantages compared to energy produced through fossil fuels (Maradin, 2021). First of all, renewable energy has almost zero greenhouse gas emissions (which we will refer to as GHG from now on) indeed, despite the fact that there are no activity-related emissions, it seems necessary to take into account the environmental impact of such a plant by going to evaluate its lifetime. Another advantage in the use of wind energy sources is manifested in the encouragement of economic development, namely the development of the energy sector and all related activities related to this industry. Finally, it seems necessary to highlight how renewable energy is outlined as a potential driving sector for rural areas, especially those underdeveloped, where this sector can contribute to their economic development. It also appears to point out how wind energy has several problematic issues related to its nature, such as:

- Because of its nature, wind energy is highly dependent on the geographical location of the plant and environmental conditions, which due to their own volatile and unpredictable nature appear difficult to estimate and with lower rates of accuracy.
- Wind energy production experiences significant fluctuations throughout the day that do not allow for a secure supply of energy throughout the day and especially during times of peak demand.
- To date, modern wind turbines have an efficiency coefficient of about 50 percent of the available kinetic energy which means that only half of the energy present is actually harnessed and converted into electricity. It also appears to highlight that, according to Betz's law (which we will discuss later) the limit of exploitation of available kinetic energy waits at 59% which would still imply that a large portion would be "wasted".

In the Graph below (Figure 2), it is possible to observe the efficiency of different technologies in electricity generation (in percent)

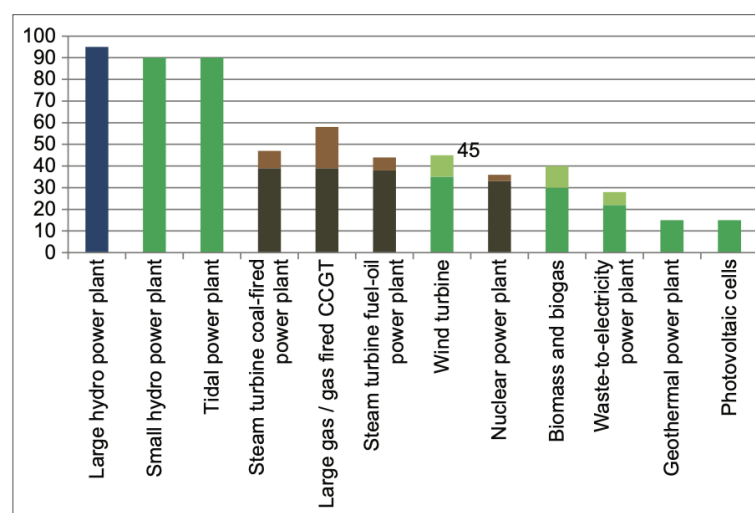


Figure 2: Efficiency of different technologies in electricity generation.
(Source: Honorio *et al.*, 2003)

- Another disadvantage of wind energy is the area required to install the plants, which consequently increases the cost of energy production (as we will see in the comparison in Figure 3). However, it seems necessary to point out that by including the costs of trading carbon dioxide emissions, wind energy source become competitive with fossil conventional energy plants (as we will be able to see in Figure 4).

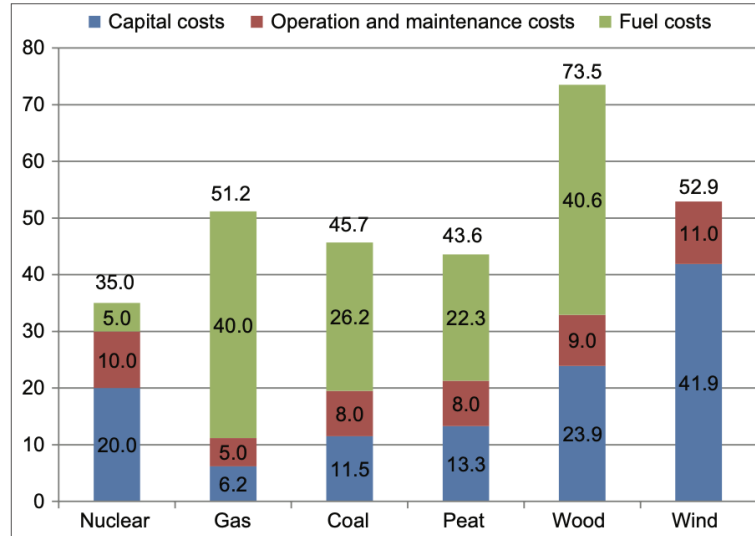


Figure 3: Electricity generation costs of different power plants and their structure, without emission trading (€/MWh).

(Source: Tarjanne and Kivistö, 2008)

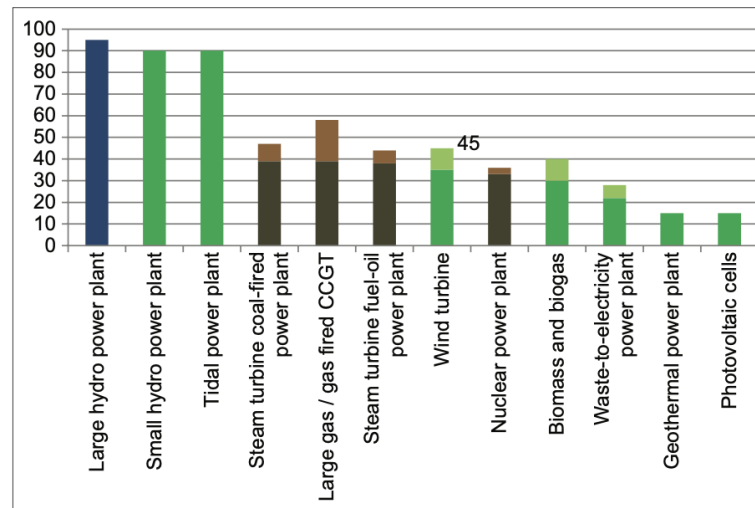


Figure 4: Electricity generation costs of different power plants and their structure, with carbon trading costs (€/MWh).

(Source: Tarjanne and Kivistö, 2008)

2.3 Forecasting models for wind power generation

Wind speed forecasting (and consequently wind production forecasting) is a very complex type of forecasting given the unpredictable nature of natural phenomena (Wang *et al.*, 2011).

During the past decades, the development of wind energy has required the development of methodologies for calculating these variables that are functional for investment evaluation. First of all, it seems necessary to clarify how forecasts are divided into 3 types in relation to the time horizon they seek to predict:

- Short-Term: it tries to predict wind speed over a time horizon from 1 hour to 72 hours ahead. This type of forecasting is critical in power system planning, for power trading in some markets where wind power and storage may be traded or hedged, for unit commitment and dispatch.
- Medium-Term: it try to predict wind speed over a horizon of 3 to 7 days. These forecasts are used in the organization of maintenance works concerning wind farms and grid.
- Long-Term: it try to predict wind speed over a horizon of 10 days to several decades. These models are mainly used for financial contracting and hedging purposes (Szczupak *et al.*, 2014).

It should also be specified that different models that can be classified into three different categories can be used to make forecasts regarding wind speed (Foley *et al.*, 2012), namely:

- Meteorological models: these models, based on historical records of temperature, humidity, atmospheric pressure, wind speed and direction, and topographical features of the area where the site under analysis is located, solve physical equations to simulate the state of the atmosphere and provide forecasts from a few hours to ten days.
- Statistical models: these models are based on time series and, through the identification of relationships between variables, are able to predict wind speed over a defined time horizon within the modeling process.
- Machine learning models: these models, through the use of advanced algorithms that train on "train datasets" and verify their learning on "test datasets", identify relationships between variables and are able to make wind speed predictions (Pelletier *et al.*, 2016).

It seems necessary to clarify that, beyond the goodness of the model itself, the accuracy of the predictions will gradually decrease as the time horizon that is being attempted to predict increases.

Chapter 3

The Data

The dataset chosen to make forecasts of wind production over the next decade was wind speed (expressed in m/s) with hourly frequency and measured at the height of the selected plant from January 1, 2014 to January 1, 2024. This frequency and period were chosen so that they could be used:

- To assess the seasonal component (which we will be able to observe later) at both the annual and infra-annual levels, while having the ability to capture seasonal fluctuations at the monthly and daily levels as well.
- To have a functional number of measurements to make the forecasts more accurate

The data was downloaded using the open-meteo API, a service that allowed data to be requested from the ERA5 dataset (a database recording different environmental variables at different altitudes and regions). In addition to the wind speed, discussed above, the temperature (evaluated at the height of the plant) expressed in Kelvin was also used in order to calculate the air density at the desired height.

3.1 Betz Law

Since the production measurements of a wind power plant are sensitive data whose detail is only available to the plant owner, it was necessary to apply different methodologies for calculating historical productions in order to continue with the analysis. To this purpose, the first methodology used was the calculation of production using Betz's law. Betz's law is named in honor of German physicist Albert Betz, who theorized it in his 1919 study entitled "Wind Energy and its Exploitation by Windmills." The above law, now considered a fundamental theoretical basis for the development of modern wind power systems, expresses the relationship between available kinetic energy and the mechanical energy that can be extracted from a rotor through the rotation of the turbine shaft.

According with the Betz limit, the wind power that can be extracted from the rotor is equal to:

$$P = C_p \cdot \frac{1}{2} \cdot \rho \cdot A \cdot V^3 \quad (3.1)$$

Where:

- P is the wind power extracted from the wind turbine rotor (mechanical power generated)

- C_p is the power coefficient, which represents the efficiency of the rotor in capturing wind kinetic energy which, according to Betz's law, cannot be more than 59.3% of the available kinetic energy
- ρ is the air density (kg/m^3)
- A is the cross-sectional area traversed by the wind flow (m^2)
- V is the wind speed (m/s).

The unit of measurement in which the power produced will be expressed will be Watt (W) since the units of measurement of the various components of the formula can be simplified as follows:

$$\begin{aligned}
 P &= C_p \cdot \frac{1}{2} \cdot \frac{\text{Kg}}{\text{m}^3} \cdot \text{m}^2 \cdot \frac{\text{m}^3}{\text{s}^3} \\
 &= C_p \cdot \frac{1}{2} \cdot \frac{\text{Kg}}{\text{m}} \cdot \frac{\text{m}^3}{\text{s}^3} \\
 &= C_p \cdot \frac{1}{2} \cdot \text{Kg} \cdot \frac{\text{m}^2}{\text{s}^3} \\
 &= C_p \cdot \frac{1}{2} \cdot \frac{\text{Kg} \cdot \text{m}}{\text{s}^2} \cdot \frac{\text{m}}{\text{s}}
 \end{aligned} \tag{3.2}$$

Knowing that

$$N = \frac{\text{Kg} \cdot \text{m}}{\text{s}^2} \tag{3.3}$$

is equal to a Newton it will be possible to rearrange the formula as follows

$$P = \frac{\text{m}}{\text{s}} \cdot N \cdot C_p \cdot \frac{1}{2} \tag{3.4}$$

that is equal to Watt (W).

Knowing then that the power will be expressed in Watt and having the data (expressed in m/s) of the wind speed it was necessary to know the dimension of the rotor (parameter A). To this purpose, through consultation of the available documentation provided by the manufacturer of the Ecotècnia ECO 80/1670 rotor, it was possible to know the size of the rotor in the site under consideration (equal to 5027 m^2). In order to know the value of the air density for each hour at the altitude of the plant under analysis, it was necessary to calculate it through the use of the ideal gas equation applied to air by correcting the value for altitude, which is equal to:

$$\rho = \frac{p}{R \cdot T} \tag{3.5}$$

Where:

- ρ is the air density (kg/m^3)
- p is the absolute air pressure expressed in Pascal (Pa)
- R is the specific gas constant for dry air, which is $287.058 \text{ J/(kg} \cdot \text{K)}$
- T is the temperature expressed in Kelvin (K)

The average annual energy produced (per plant), according to Betz's formula, was 2700MW which is 400MW higher than stated in the latest reports released by the site owner. This difference is attributable to the variable C_p (i.e., the power coefficient). In fact, this information appears to be unavailable and not otherwise calculable

3.2 Energy production by using power curves

Given the discrepancy between the calculated (average) values and the values provided by the wind farm owner the second method used to estimate the energy produced was to use the power curve related to the above-mentioned Ecotècnia ECO 80/1670 rotor (whose graphical representation is given to you below in Figure 5). Obviously, finding wind speed values not perfectly equal to the power curve values, a process of interpolation of the curve was first necessary (Lydia *et al.*,2014).

The theoretical formula for linear interpolation is an expression that, given two points (x_1, y_1) and (x_2, y_2) where x_1 and x_2 are the wind speeds and y_1 and y_2 are the corresponding powers, returns the value of y corresponding to a certain wind speed x .

The mathematical formulation is the following:

$$y = y_1 + \frac{(x - x_1) \cdot (y_2 - y_1)}{x_2 - x_1} \quad (3.6)$$

Using this calculation methodology, the discrepancy between the average annual production values provided by the plant owner and those calculated was about 100MW (difference attributable to the difference in the time horizon calculated for this study and the time horizon over which the plant owner averaged).

Consequently, given the better fit of the production values calculated through power curves, we will use these values as historical data from now on.

Power curve

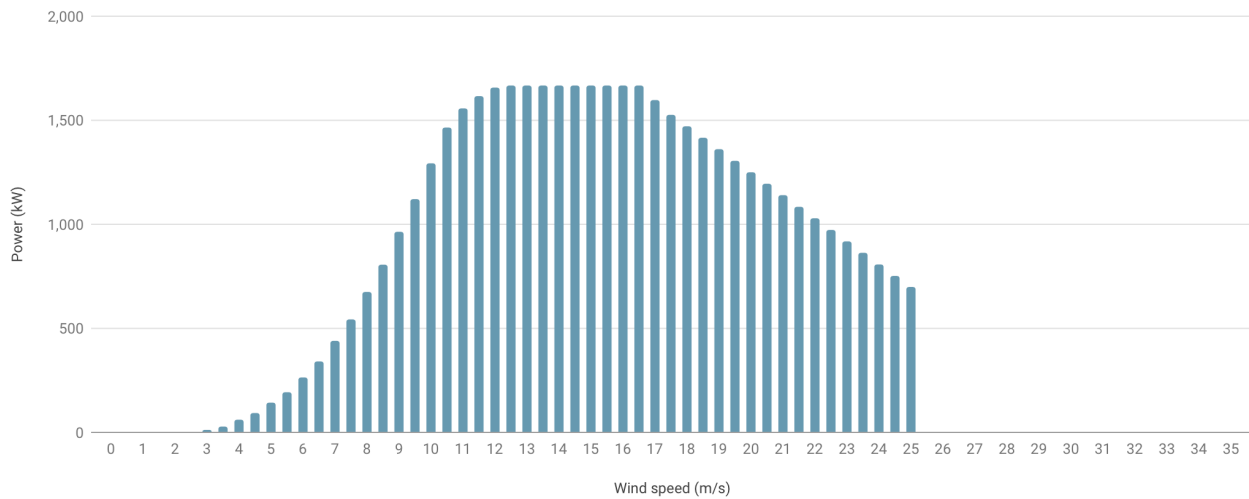


Figure 5: Power curve of the Ecotècnia ECO 80/1670 rotor. (Source: wind-turbine-models.com)

3.3 Historical production data analysis

The objective of the following section is to analyze and envisage the characteristics of the historical data obtained through the above calculations. In order we will first go to examine the Weibull distribution, which will be shown for the historical data and will later be used for forecasting purposes, then the trend and seasonality components of the historical series will be analyzed as well as the main statistical measures.

Finally we will be able to see graphs regarding historical production in order to appreciate the variations during the different months of each year under consideration and the overall trend during the period under analysis.

3.3.1 Weibull

The Weibull distribution is a probability distribution, described by the Swedish mathematician Weibull, particularly used to model events involving a time dimension and widely used in modeling the probability of wind speed. The formula of the Weibull distribution is defined as follows:

$$f(v) = \frac{k}{\lambda} \left(\frac{v-c}{\lambda} \right)^{k-1} e^{-\left(\frac{v-c}{\lambda}\right)^k} \quad (3.7)$$

Where:

- v is the random variable, representing the wind speed in our context.
- k is the shape parameter of the Weibull distribution. It determines the shape of the distribution: higher values of k correspond to a longer right tail and a spikier shape, while lower values of k produce a shorter right tail and a more rounded shape.
- λ is the scale parameter of the Weibull distribution. It determines the horizontal location of the distribution along the axis v . Higher values of λ indicate a wider distribution, while lower values indicate a narrower distribution.
- c is the location parameter indicating the horizontal shift of the distribution.

In Figure 6, the historical wind speed readings adhere almost perfectly to the Weibull distribution. The obtained parameters (equal to $k = 1.45$, $c = 0.09$ and $\lambda = 4.36$) confirm a moderately sharp shape of the distribution. The beginning of the distribution at the point $x = 0.09$ and an extent of the distribution equal to 4.36 indicates that in the proximity of this value there is the highest concentration of measurements.

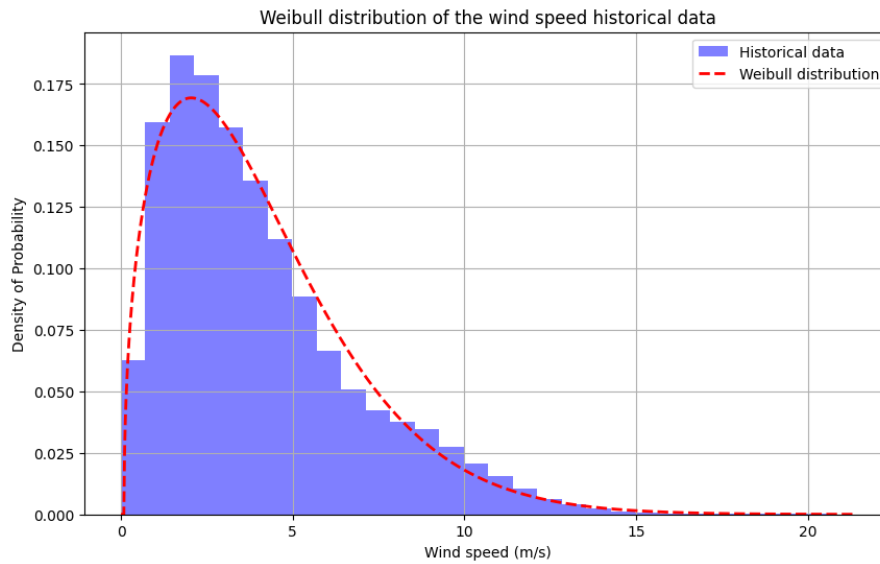


Figure 6: Weibull distribution and historical data

3.3.2 Seasonality

Wind, like other natural elements, suffers cyclical fluctuations. Such fluctuations are identified as seasonality. Thus, with seasonality we identify all those cyclical variations that occur over a certain period of time.

The above variations can be influenced by a number of factors, such as:

- Seasonal effects: we are talking about seasonal effects since the seasons of the year can affect wind speed.
- Diurnal cycles: during the course of a day, wind speed may be subject to fluctuations caused by variation in solar radiation, atmospheric convection, and atmospheric stability
- Meteorological phenomena: wind speed is also subject to cyclical weather phenomena such as tropical cyclones and atmospheric tides.

Aware of the alteration of wind speed therefore, it was needed to seasonally adjust the historical series in order to better study its components without being affected by cyclical fluctuations of the lowest forecasting relevance. For this purpose, it was necessary to perform a Seasonal and Trend decomposition process using Loess (which we will refer to as STL) that allowed us to obtain:

- The trend component: indeed, through the Loess method a local regression model is fitted to each data point, weighting the closest points more heavily than the farthest points. This allows nonlinear or complex patterns to be captured.
- The seasonal component: this component is calculated by applying Loess method to the residuals obtained during the trend calculation allowing the model to capture cyclical or periodic patterns not explained by the trend component.
- the residual component: following the calculation of the trend and seasonal component, residuals are derived by subtracting the sum of the trend and seasonal component from the original

data. The residuals thus represent the random fluctuations in the data that cannot be explained by the other two components.

The first result obtained through STL that was analyzed graphically (and shown in Figure 7) was the trend, seasonality and residuals of the entire time series related to wind speed.

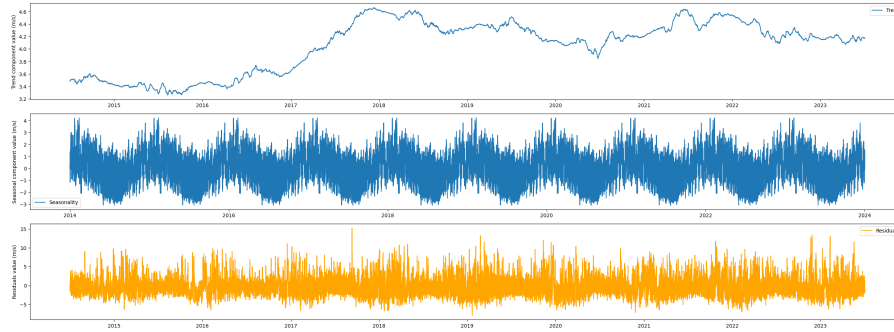


Figure 7: Trend, Seasonality and Residuals of the whole timeseries

It is immediately noticeable from the graphical analysis how the trend has been growing strongly since the second half of 2016 and then stabilizing at a less variable value from 2018 until the last survey included in the time series. This can be explained by the increasing temperature, which, as explained earlier, influences the wind speed. From the second graph in the figure, it can be seen that the historical wind data are strongly affected from the seasonal component which takes a sinusoidal shape. The value of seasonality represented takes values in the range of -3 to +4 assuming its maximum values always a little more than halfway through the year and at the beginning of the following year. In any case, in order to gain a better understanding of the seasonality of the time series, I decided to also analyze the annual frequency of the phenomenon in order to understand its pattern between months.

The third graph (in yellow) shows the residual component that cannot be explained by trend and seasonality. This graph shows us a residual component between -5 and 15 but with a strong concentration between -1 and 1.

As anticipated earlier, in order to learn more about the structure of seasonality and residuals the STL process was applied by going to consider each year separately in order to know in greater detail the length (in months) of the seasonal cycles within each year. Indeed, the figure below shows, for each year, the seasonal component, and the residuals ones.

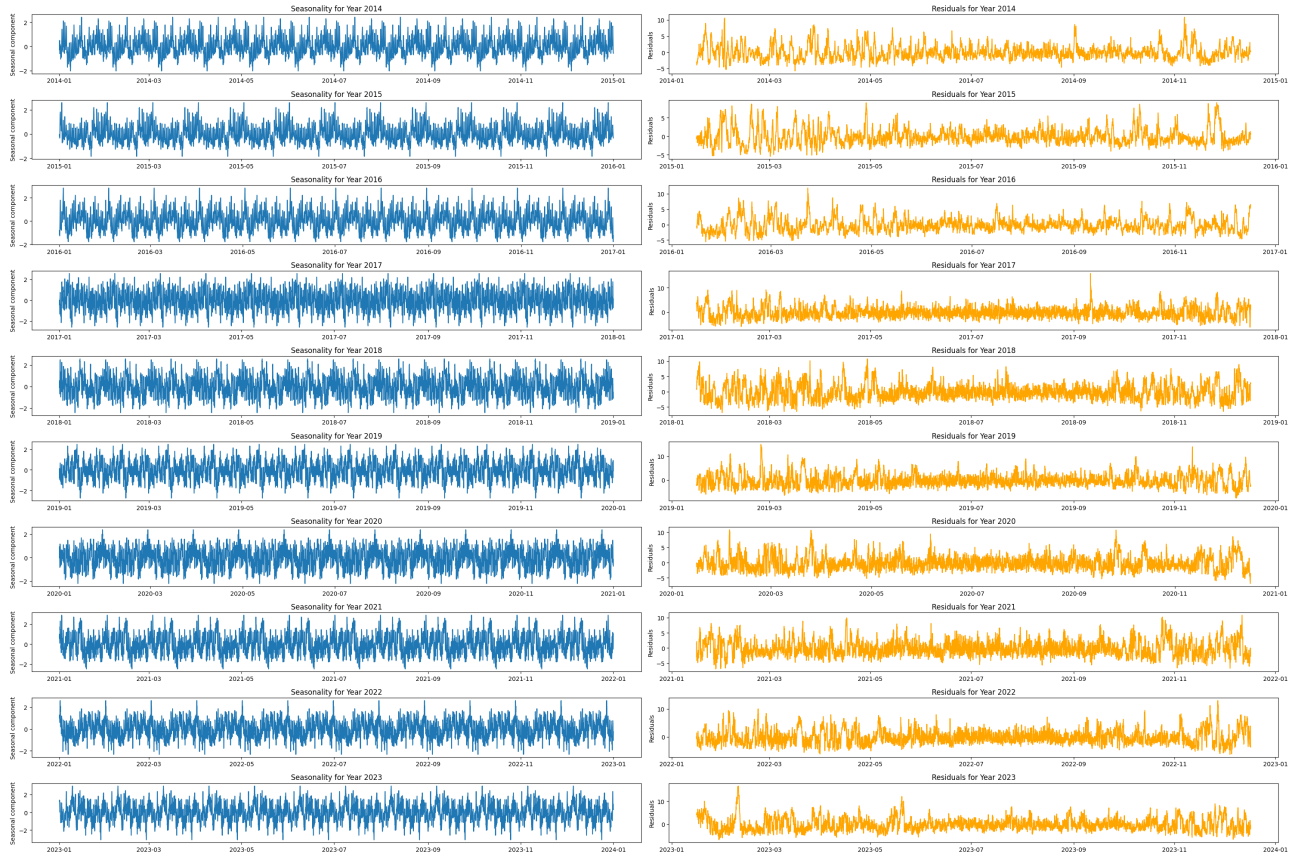


Figure 8: Annual Seasonality and Residuals component

The analysis in Figure 8 shows a clear cyclical connotation even at the annual level it shows cycles of bimonthly amplitude. It also seems relevant to note that over the years seasonality has varied, which can be attributed to:

- Changes in local topography: such changes, such as urbanization or deforestation in surrounding areas, can alter the shape of the terrain and reflective surface, thereby altering wind speed.
- Long-term seasonal variations: in fact, as explained earlier, increasing temperature could be a reason for such alteration of annual seasonal patterns.
- Human activity: this aspect particularly concerns all polluting activities, since it can have direct and indirect effects on cloud formation and atmospheric stability. These are factors that, in turn can alter wind patterns.

Clearly, wind production being a function of wind speed will also follow the same trend followed by wind speed and be subject to the same seasonal fluctuations.

3.3.3 Energy produced and statistical measures

After being able to observe the different wind components by means of Seasonal-Trend decomposition using Loess and being able to observe the probabilistic distribution of wind speeds, the following section will show how, using the above power curve, historical wind production was calculated.

As previously mentioned, the first step in calculating the energy actually produced by the wind farm was to interpolate its power curve. For this purpose, first of all, it was necessary to define the

power curve and go to distinguish its different speed-production pairs. Next, the interpolation process was defined by constructing the function paying special attention to the observance of the upper and lower bound, corresponding respectively, to 25 m/s, i.e. the speed above which the plant shuts down, and 3 m/s, i.e. the speed below which the plant does not turn on. Thus, having the interpolated powercurve for all possible wind values between the limits, it was necessary to define a second function needed to calculate the energy produced. By doing so it was possible to identify the amount of energy produced for each hour of the wind speed timeseries. By aggregations on different time scales it was possible to obtain the following graphs.

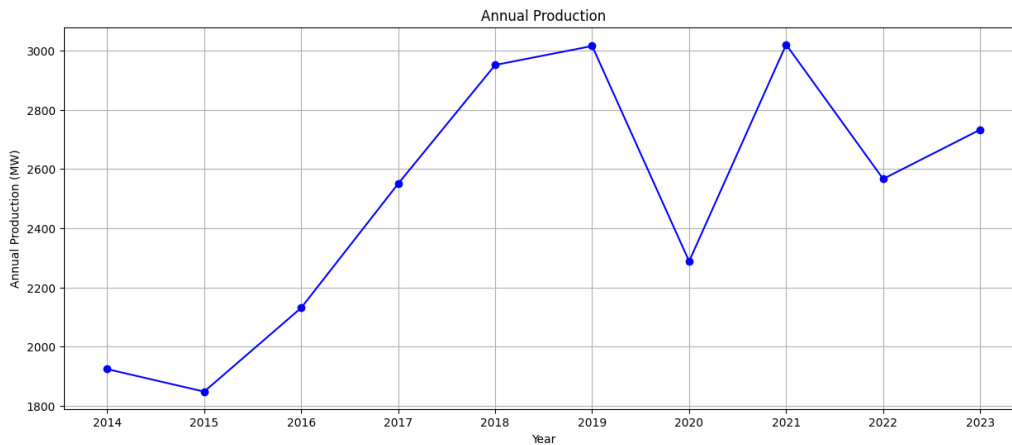


Figure 9: Historical wind power production (10 years)

In Figure 9, it is possible to observe the different amount of wind energy produced in each of the past 10 years, also it is possible to recognize a development of the graph similar to that found previously when the trend was calculated (Figure 7). Subsequently, in order to visually analyze and be able to better understand the intra-annual variations, it was also necessary to perform the graphical representation (in the figure below) of wind production separately for each year, thus knowing its monthly patterns.

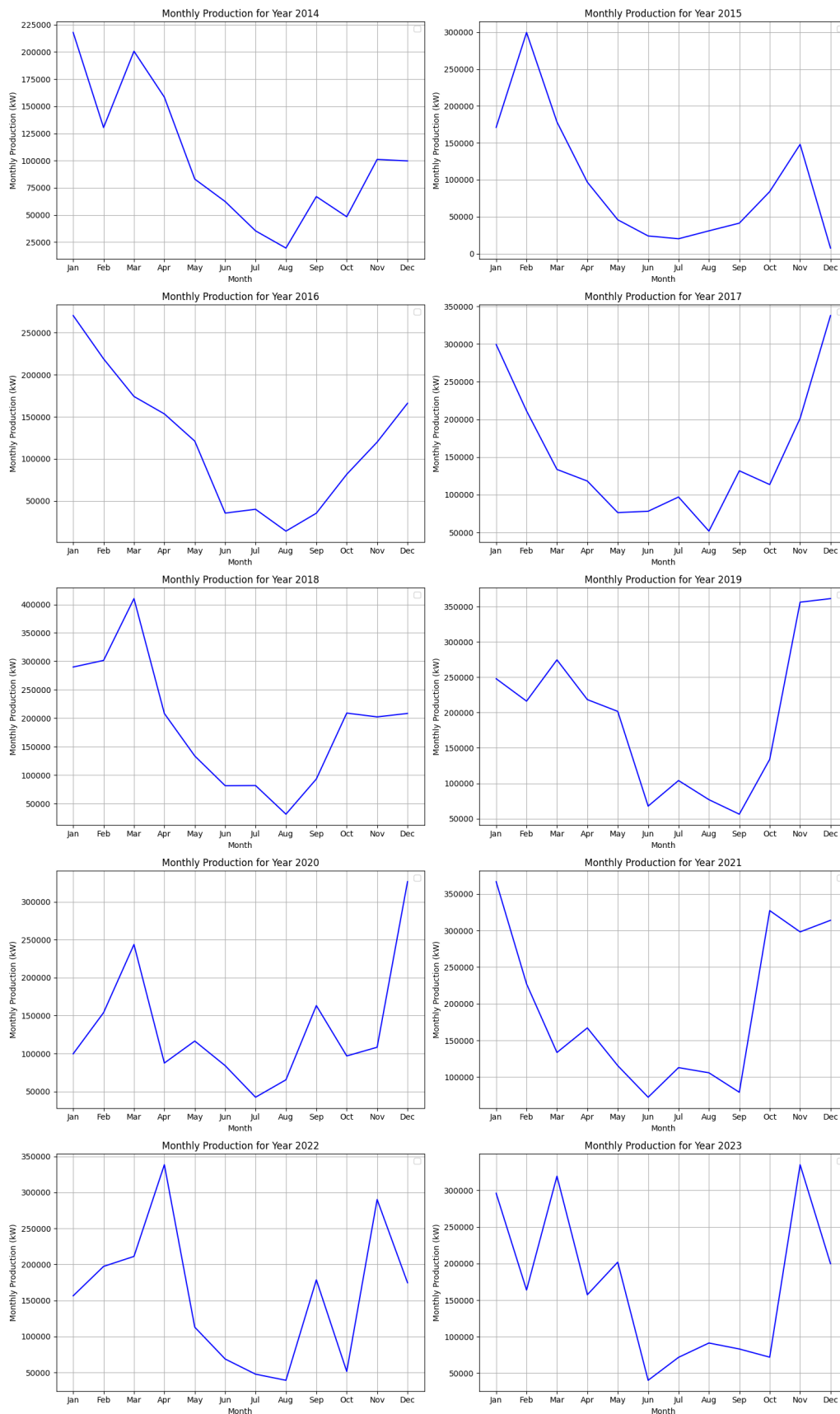


Figure 10: Annual wind power production

From the figure it is possible to apprehend that there is increased production during the first and last months of each year, with values almost always reaching a maximum around March and then decreasing reaching minimum values during the summer months (June-August) and then beginning to increase again until reaching new peaks around November.

The last step in the analysis of historical production results was to evaluate the main statistical measures shown in the table below.

Mean	Variance	Standard Deviation	Kurtosis	Skewness
205.67268459558574	148839.7836963778	385.7975942076075	5.002556615965608	2.389529345088855

Figure 11: Historical wind energy production: main statistical measures

It can be understood from this table that:

- The average hourly production appears to be equal to 205.67 kW
- The variance of the data, corresponding to the variability of the measurements from the average, appears to be very high and equal to 148839.78. This appears to be due to the fact that, below 3 m/s of wind speed, the production will be equal to 0 due to the plant not being turned on. Furthermore, in cases where there has been wind above 25 m/s, the production will be equal to 0 since the plant will go into shutdown
- The standard deviation, by using the same scale as the calculated values (corresponding to kW) clarifies how the values differ by 385.7975 from the mean. This appears to be in line with what has been elucidated regarding variance since the standard deviation corresponds to the square root of the variance itself.
- Kurtosis is the statistical measure that indicates the shape of the distribution, and, a value of 5, such as that shown in the table, indicates that the distribution has much heavier tails than a Normal.
- Skewness measures the skewness of the distribution and, a value of 2.38 indicates a lengthening of the right tail of the distribution, thus confirming the permanence of the Weibull distribution even with regard to the energy produced

Chapter 4

Forecast

In the following section, we will take an in-depth look at various forecasting models, initially from a theoretical point of view and then from a practical one, to understand the steps and logic involved in executing the forecasts. First, it seems necessary to mention that, as also expressed earlier, the time horizon appears to be a feature strongly impacting the accuracy of the model. All models, because of the cumulative prediction errors for each forecast step, will exhibit more accurate forecasts for very short-term time horizons (in the case under analysis equivalent to hours), and increasingly uncertain forecasts as the time horizon increases. The models used, which we will see in detail later, are based on two different logics:

- Autoregressive models: such models are used to model and predict time series. The goal of the model is to identify correlations between historical data and, through these linear relationships, predict future trends.
- Models based on probability distribution: these models aim, through analysis and modeling of the historical probability distribution, to make predictions about possible future patterns.

In the modeling performed, in order to compare the results, both types of models were run, and, in particular, an Autoregressive Integrated Moving Average (ARIMA) was used as the autoregressive model and a Monte Carlo simulation as the probabilistic model.

4.1 Monte Carlo Simulation

Simulation is a type of modeling used in the study of the behavior of a system (to be considered in its broadest sense) through its reproduction in a controllable context. In simulation, the relationships between the components of the system under analysis and how they affect its behavior are defined in order to perform sufficiently accurate simulations to study the system's behavior. This type of analysis appears to be of fundamental importance in order to:

- Increase the understanding of the operation of the system itself
- Being able to check hypotheses
- Being able to make predictions

Monte Carlo simulation is a technique used to reproduce and solve numerically problems in which random variables are also involved, the analytical solution of which would otherwise be excessively complex or impossible (Yun *et al.*, 2021). Furthermore, through the use of the above simulation, it appears possible to modify the input characteristics of the model in order to learn about the behavior of the system under conditions that are different from historical ones.

Monte Carlo's method appears to have been employed as early as the 1700s, as evidenced by the work of the French mathematician George Louis Leclerc de Buffon, who described an experiment in estimating the value of π by a random simulation based on the throwing of a pin, obtaining results that were quite accurate for the time.

The Monte Carlo's method then appears to have been used extensively in nuclear research in the 1900s being able to rely, moreover, on electronic calculators since the 1940s. Today, this method is used in various scientific fields and, in particular, in investment evaluation. This application is probably due to David Hertz who, in his article *Risk Analysis in capital investment*, proposed this technique for the evaluation of a chemical plant expansion project.

The components of the Monte Carlo's model can be described as:

- The parameters: i.e., inputs specified by the analyst that are therefore controllable
- The input variables: these correspond to exogenous characteristics that depend on events not under the control of the analyst, whose probabilistic characteristics can therefore only be fitted
- The output variables: that is, the results of the simulation
- The model: i.e., the mathematical equations that describe the relationships among the components of the system.

After defining all the components of the model, the Monte Carlo method solves the problem numerically, i.e., by producing an N (analyst's choice but still sufficiently large) number of possible combinations that the input variables can take, and calculating the corresponding output based on the model equations.

For the generation of each of the combinations, a value is randomly generated for each input variable (in accordance with the specified probability distribution and respecting the correlations between the variables). By repeating N times this procedure, it will be possible to obtain N different values of the output variable, which represent a sample of the possible values that can be assumed by the output.

As previously clarified with the Monte Carlo's method, it will not produce an analytical formulation but rather a sample of values, the frequency of which allows us to derive an approximate indication of the distribution of the variable. Consequently, with a low number of samples it would also not be possible to calculate statistical indicators of interest (such as the mean and standard deviation).

Generating a large number of samples allows for greater precision and accuracy since this would in practice represent an application of the central limit theorem. In fact, the central limit theorem states that, under the assumptions of the law of large numbers, that is, in the presence of N independent and identically distributed random samples X_i with mean μ_x and variance σ_x^2 , the random variable $(\bar{X} - \mu_x)/\sigma_{\bar{x}}$ ((where $\sigma_{\bar{x}}^2 = (\sigma_x^2)/n$ can be approximated to the standard normal distribution).

In formula we will then have:

$$\frac{\bar{X} - \mu_x}{\sigma_{\bar{x}}} = \frac{\bar{X} - \mu_x}{\frac{\sigma_x}{\sqrt{n}}} \sim N(0, 1) \quad \text{for } n \rightarrow \infty \quad (4.1)$$

Where $\bar{X}$ is the sample mean defined as follows:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \quad (4.2)$$

Consequently, it is possible to say that this result is approximately true when n is sufficiently large even if $X_1, X_2, \dots, X_n$ are not normally distributed.

4.1.1 Monte Carlo Modeling

In the following section we are going to look at the specifications used in modeling a Monte Carlo simulation to perform a probabilistic forecast of (hourly) wind speed values over the next 10 years. The first step was to define within the model all the parameters of the Weibull distribution that are:

- k : The shape parameter equal to 1.45 in the wind distribution under analysis
- λ : The scaling parameter, equal to 4.36 in the specific case analyzed
- c : The location parameter, equal, according to the calculated distribution, to 0.092

The above parameters have the function of defining the shape and scale of the distribution, consequently affecting the variability and range of wind speed values.

Through the use of the parameters, it was possible to generate the different Monte Carlo paths to simulate the wind speed in each hour for the next 10 years. Each path obtained therefore represents a time series of wind speed values sampled from the Weibull distribution.

For the reasons explained earlier, it is necessary to generate a large number of simulations. Empirically it has been shown that already 30 simulations appear to be an adequate number to comply with the central limit theorem, but with the aim of obtaining a representation as accurate as possible, 100 different paths were computed in the present model.

Following the calculation of the 100 different paths, it was considered necessary to observe the graph (depicted in the Figure 12 below) of all the simulated paths in order to be able to verify their characteristics.

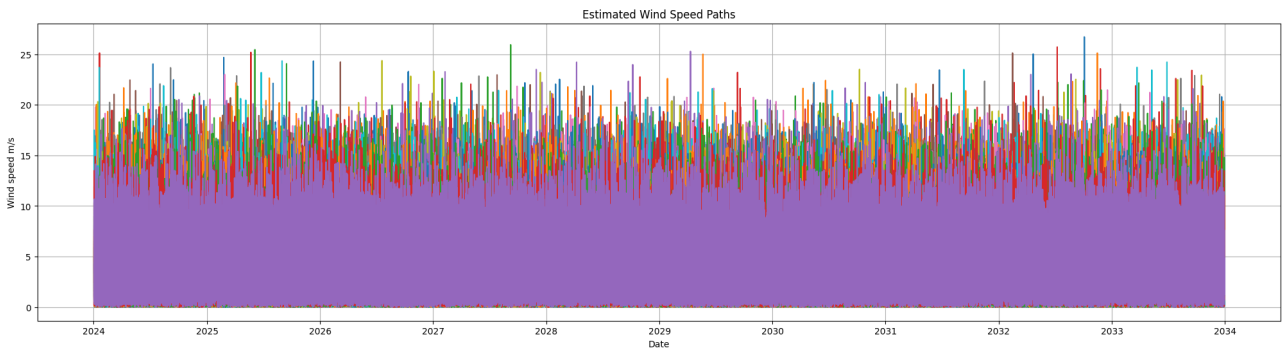


Figure 12: Monte Carlo simulations of hourly wind speed (10 years ahead)

As can be observed from the Figure 12, each of the different colored lines represents a different path calculated by simulation. It can also be observed from the graph that only some paths reach 25 m/s, which is the maximum value. This is due to the different path generation but is at the same time an indication of the need of path manipulation in order to perform a more accurate estimation of the

energy produced. Therefore, in order to make a future forecast as accurate as possible and to eliminate deviations caused by outlier values, as clarified by the theory of the Monte Carlo method, it was necessary to calculate the average path, i.e., the path that for each hourly forecast had as its value the average of all hourly forecasts calculated over all simulations.

The path calculated by such data manipulation appears to be as shown below in the Figure 13.

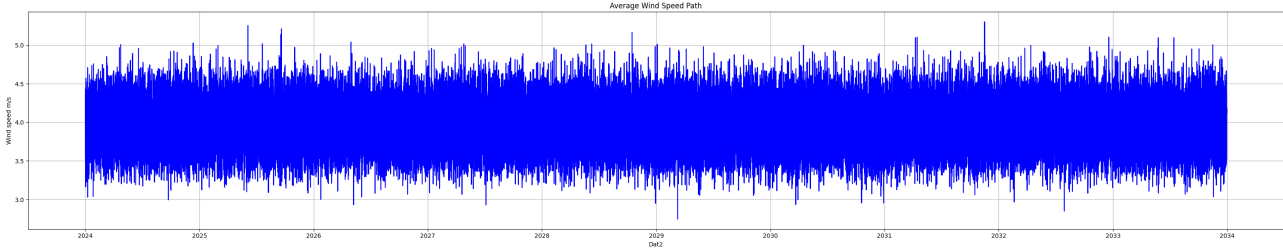


Figure 13: Average Monte Carlo path of hourly wind speed (10 years ahead)

As can be seen immediately by looking at the graph this appears to have less variability among the observations and more concentration around the mean value. This is due to the nature of the mean itself and the law of large numbers, in fact:

- **Mean and variance:** when the mean of a set of values is calculated, a central representation of them is obtained in fact, by definition, the mean tends to “smooth out” extreme variations since this is the result of aggregating all available data
- **Law of Large Numbers:** this law states that as the number of observations increases, the average of them tends to converge to the expected value. This means that, given a sufficient number of Monte Carlo simulations, the average of these paths will be very close to the expected value of the distribution from which it is being sampled (in the case under analysis a Weibull distribution)
- **Variance reduction:** by calculating the average of a large number of paths the random fluctuations (corresponding to the variance) of each path tend to cancel each other out. In fact, considering σ^2 as the variance of a single Monte Carlo path, then the variance of the average path will be equal to:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \quad (4.3)$$

Where $\bar{X}$ identifies the mean path of the simulation.

- **Concentration around the mean:** since the variance of the mean path decreases as the number of paths increases (for the reasons already discussed), it follows that the mean will be less subject to extreme variation and more concentrated.

It is therefore possible to say that the variations that occurred as a result of calculating the mean path appear entirely consistent with the very nature of the concepts of variance, mean and the law of large numbers. Moreover, such a result ensures that the observation is stable and representative of the expected value.

4.1.2 Results

After performing all the above steps, it was necessary to calculate the estimated wind power production based on the forecasts obtained. The power curve discussed above was used for this purpose, and in order to calculate the energy produced for any future wind value, it was necessary to perform the curve interpolation process again. The figure below then shows the graph representing the values (expressed in kW) of the energy produced for the upcoming periods.

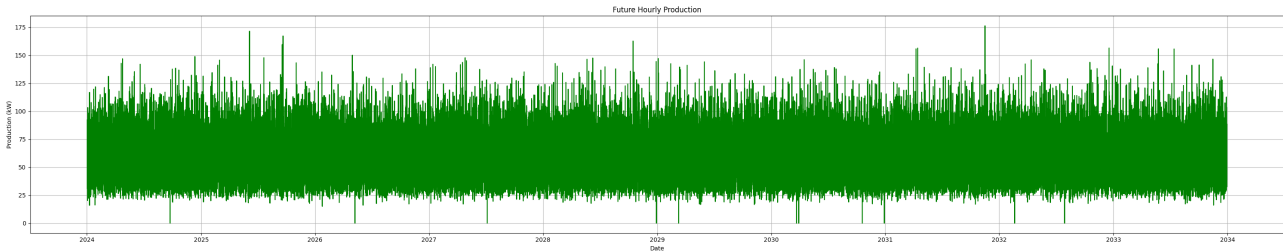


Figure 14: Wind production calculated by average Monte Carlo simulation (10 years ahead)

It can be seen from the Figure 14 that the amount produced is highly concentrated around the mean, and again, this is due to the smoothing effect obtained by calculating the mean path.

Finally in the table below we can observe the main statistics regarding the energy produced.

Mean	Variance	Standard deviation	Kurtosis	Skewness
61.36839662684335	326.0776874511731	18.05762131209903	0.3994575367270876	0.34440886181822183

Figure 15: Predicted wind energy: main statistical measures

It can be seen from the measures shown in the table that:

- **Mean:** this measure indicates the calculated average hourly production value, which, in the case under analysis, is found to be approximately 61.37 kW
- **Variance:** this measure indicates the degree of variability of the readings from the mean. As this value increases, the variability of the measurements from the mean will also increase
- **Standard Deviation:** this measure corresponds to the square root of the variance and allows to quantify the distance from the average through the use of the same scale as the original data, so that, as for the case under analysis, we can see that future values of hourly production differ by about 18.06 kW from the average.
- **Kurtosis:** this measure indicates the shape of the distribution of the data. A positive kurtosis indicates that the distribution has heavier tails than the Normal distribution, while a negative kurtosis indicates already light tails. In the case under analysis, the value of 0.399 suggests that the distribution has slightly heavier tails than a Normal
- **Skewness:** this measure assesses the symmetry of the distribution of the data with respect to the mean. In the case under analysis we can see a value of 0.344 which indicates a slight positive skewness that, graphically, indicates that the tail of the distribution is slightly more elongated on the right side of the mean than on the left side

4.2 Time series analysis

The following section will describe from a theoretical point of view the operation and meaning of a moving average autoregressive model.

First, it is necessary to clarify that such models can be used in the analysis of time series (Schulz *et al.*, 2001). Time series are composed of records taken at regular intervals of one or more variables. Based on this characteristic it is possible to distinguish two different categories of time series:

- Univariate timeseries
- Multivariate timeseries

In the univariate case the variable under observation depends solely on the past values. In contrast, in the multivariate case the dependence extends to the other variables as well.

Multivariate time series are derived from complex phenomena that cannot be described by univariate models, since these lack the ability to capture dependencies among the different variables that make up the time series itself. Regarding timeseries it should also be specified that they can be decomposed into a finite number of components that are not directly observable, corresponding to:

- The trend: that is, the development of the long-term series, this can be increasing, decreasing or stable and can change its slope over time
- Seasonality: corresponding to cyclical patterns affecting the data. This component can occur in different frequencies in the timeseries.
- White noises: corresponding to all variations not belonging to the previously described components and having a random nature.

A relevant property of the time series is that, in general, adjacent observations appear to be dependent, and this feature takes on fundamental relevance for analytical purposes.

In fact, through the use of the observations available at time t , it is possible to predict the value at a future time defined as $t + l$ (where l corresponds to the lead time, i.e., the time between the beginning of the process and its conclusion). Obviously, the time horizon for which the forecast calculation is performed is variable and depends on the nature of the phenomenon one is trying to model, the purpose of the analysis, and the frequency and size of the historical time series.

It is necessary to clarify, however, that first of all, in order that such a modeling is possible, it is necessary for the observations to be defined in a discrete time interval and for them to be equidistant (i.e., all have the same frequency). Models used in the calculation of time series forecasts are called stochastic because, since probably no phenomenon is totally deterministic, it is not possible to define a model that exactly calculates the future behavior of the variable under analysis having an initial phenomenon also influenced by different unknown factors. An important category of stochastic models for describing time series includes stationary and nonstationary models.

- Stationary models assume that the process remains in statistical equilibrium with probabilistic properties that do not change over time, and that vary around a fixed constant mean level and have constant variance
- Non-stationary models are those models in which there is variation in the characteristics of the underlying stochastic process.

4.2.1 Backshift operator

Before continuing with the examination of the models, it needs to explain the concept of the backshift operator. This operator, also called “lag” in the literature, applied to the variable of interest has the effect of taking the data back a number of periods defined by the analyst. The mathematical formulation of such an operator corresponds to:

$$B^x y_t = y_{t-x} \quad (4.4)$$

Where:

- B is the symbol for the backshift operator
- x corresponds to the “lag” number, i.e., the past period being examined
- y_t corresponds to the variable being analyzed at the current time

The backshift operator plays a key role in model definition by allowing the differentiation process to be performed in the case of nonstationary timeseries.

4.2.2 Differentiation

As previously explained time series can be stationary and non-stationary, but as long as it is possible to apply autoregressive (AR), moving average (MA) or a combination of the two (ARIMA, ARMA, SARIMA) models, it is necessary to manipulate the series in order to make it stationary.

This operation is known as differentiation. The differentiation of a time series can best be understood from the perspective of mathematical differentiation. In simple terms, differentiation corresponds to calculating the instantaneous variation of a series with respect to time, that is, it involves calculating the discrete derivative of the series with respect to time.

Mathematically, if $y(t)$ is a timeseries, the finite difference between two consecutive points can be expressed as:

$$\Delta y(t) = y(t) - y(t - 1) \quad (4.5)$$

Where:

- $\Delta y(t)$: Correspond to the discrete derivative at time t
- $y(t)$: Correspond to the value of time series at time t
- $y(t - 1)$: Correspond to the value of time series at time $t - 1$

As mentioned, differentiation aims to make the time series stationary, but, however, one differentiation is not always sufficient. Consequently, one proceeds iteratively to calculate the difference of differences.

Differentiation orders are divided into three main types:

- First-order differentiation: corresponds to the case, presented in formula above, in which the difference between the value of the series at time t and the value at the previous time $t-1$ is calculated.

- Second-order differentiation: if the series is not stationary after the first differentiation is needed to proceed in repeating the calculation but, this time, on the differences obtained from the first differentiation.

The mathematical formula is thus:

$$\Delta^2 y_t = \Delta(\Delta y_t) = \Delta y_t - \Delta y_{t-1} \quad (4.6)$$

- Higher orders: if even after the second differentiation the series still appears non-stationary it will be necessary to repeat the differentiation process n times, until the series is stationary.

The mathematical formula then appears to be equal to:

$$\Delta^x y_t = \Delta(\Delta^{x-1} y_t) = \Delta^{x-1} y_t - \Delta^{x-1} y_{t-1} \quad (4.7)$$

In conclusion, it seems important to mention that the differentiation process appears to be the most suitable methodology to make the time series stationary and thus analysable by stochastic models, but at the same time the differentiated series may show more variability or white noises, especially if the order of differentiation is high.

4.2.3 Invertibility

In time series analysis, invertibility appears to be a fundamental concept concerning the ability of a model to represent each value in the series as a linear combination of its past values and current and past residuals. In other words, an invertible model allows the time series to be reconstructed from its residuals and past values. In the context of stochastic models, invertibility appears to be an essential condition to ensure that the model is suitable for time series analysis and forecasting.

To consider the invertibility of a model we consider a general time series process, equal to:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} \quad (4.8)$$

Where:

- y_t : corresponds to the value of the time series at time t
- ϕ : correspond to the autoregressive coefficients, which will be defined in depth later
- ε : Corresponds to the white errors of the timeseries
- θ : corresponds to the moving average coefficients of the process

For such a model to be invertible, all $\theta_1, \theta_2, \dots, \theta_q$ coefficients must be such that the absolute value of each coefficient is less than 1. Invertibility has been defined by *Box* and *Jenkins* as one of the fundamental characteristics for time series analysis, since a model that is not invertible can lead to problems in parameter interpretation and time series forecasting, producing unstable or unreliable predictions.

4.3 Autoregressive model (AR)

A first stochastic model that may be useful for modeling certain types of time series is the Autoregressive model. This model is applied when the variable of interest is predicted by a linear combination of past values of it (in fact, the term “autoregressive” means a regression of the variable of interest on itself).

This model assumes that the present value is:

- Expressed in finite form
- Is a linear aggregation of the previous values of the process with the addition of a random component

This characteristic appears even clearer through its mathematical formulation, which is equal to:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t \quad (4.9)$$

Where:

- c : corresponds to a constant term
- y_t : corresponds to the value of the time series at time t
- $\phi_1, \phi_2, \dots, \phi_p$: correspond to the autoregressive coefficients, which are the terms that quantify the linear relationship between the current value of the time series and its past values.
- $y_{t-1}, y_{t-2}, \dots, y_{t-p}$: correspond to the past values of the timeseries
- ε_t : corresponds to the white noises (whose meaning was clarified earlier) at time t

In conclusion it is possible to say that this specific model will be referred to as “Autoregressive model of order p ” where p corresponds to the value of lags.

Calculation methods and function of coefficients ϕ : Since autoregressive coefficients play a crucial role in the model, it seems necessary to investigate the different calculation methodologies in more detail.

The first method is the Ordinary Least Squares (OLS) method. This method aims to minimize the sum of squares of the differences between the observed values and the values predicted by the models. The implementation of this methodology requires solving a system of linear equations in order to identify the ϕ that minimize the overall error. Although the OLS method appears to have a simple and intuitive implementation it is at the same time subject to lower efficiency in the presence of complex serial correlation.

The second methodology for calculating coefficients is Maximum Likelihood Estimation (MLE). The underlying logic of MLE is to identify parameter values by maximizing the probability of observing the data actually provided by the model. MLE appears to be a good method for parameter estimation since the estimates appear to be consistent and, as observations increase, the estimates converge to the actual values of the parameters. However, at the same time, it must be said that MLE requires a great deal of computational complexity and the estimates appear to be very sensitive to outliers in the data.

In general, in the case where the coefficient is positive ($\phi > 0$) then it will indicate that an increase in the past value $\{y_{t-i}$ is associated with an increase in the current value y_t .

Conversely in the case where the coefficient is negative ($\phi < 0$) then it will indicate that an increase in the past value of $\{y_{t-i}$ is associated with a decrease in the current value y_t .

4.4 Moving Average model (MA)

A second stochastic model for time series modeling is the moving average (MA) model. Such a model instead of using past values of the dependent variable as in the case of autoregressive models, uses past errors as predictors in a model whose formulation (below) appears similar to that of regression.

$$y_t = c + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q} \quad (4.10)$$

Where:

- c : corresponds to a constant term
- y_t : corresponds to the value of the time series at time t
- $\varepsilon_{t-1}, \varepsilon_{t-2}, \varepsilon_{t-q}$: it is the error term at time t . this term represents the random or irregular component of the series at the specified time t
- $\theta_1, \theta_2, \theta_q$: correspond to the coefficients of the model. Each coefficient θ_i represents the effect of the error term at time $t - i$ on the present value of the time series at time t

The methodology for estimating the Moving Average model parameters correspond to the same methods already described for calculating the Autoregressive coefficients (OLS and MLE).

4.5 Mixed Autoregressive-Moving Average Models

In the following section we are going to explore the different models that can be used through the composition of the autoregressive (AR) model and the moving average (MA) model by analyzing what are called the Autoregressive Integrated Moving Average (ARIMA) model and the Autoregressive Moving Average (ARMA) model.

4.5.1 Autoregressive Moving Average (ARMA)

An Autoregressive Moving Average (ARMA) model is a process that includes both a moving average and an autoregressive component through the practice of the additive combination of an $AR(p)$ and a $MA(q)$ process.

The general formula of such a process is the following:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q} \quad (4.11)$$

The above formula can be reformulated in terms of the two characteristic polynomials, equal to:

$$c + (1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p) y_t = (1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q) \varepsilon_t \quad (4.12)$$

Which can be simplified as:

$$\Phi(B)y_t = \Theta(B)\varepsilon_t \quad (4.13)$$

The stationarity of the ARMA process only concerns the autoregressive component since the moving average is stationary by definition. Conversely, invertibility is a characteristic that only concerns the moving average component since the autoregressive part is invertible by definition. The characteristics mentioned above imply that the ARMA process (p, q) is stationary if and only if the roots of the characteristic polynomial of the AR component $\Phi_p(B)$ are, in modulus, greater than one. Mathematically, the formula for verifying stationarity is equal to:

$$y_t = \Phi(B)\Theta_q(B)\varepsilon_t = \left[\frac{1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q}{1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p} \right] \varepsilon_t \quad (4.14)$$

Which can be rewritten as:

$$y_t = \Phi(B)\Theta_q(B)a = \left[\frac{(G_1 - B)(G_2 - B) \dots (G_p - B)}{1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p} \right] \varepsilon_t \quad (4.15)$$

Where G_i with $i = 1, 2, \dots, p$ are the roots of the characteristic polynomial of the autoregressive component. If these roots are such that $|G_i| > 1$ for $i = 1, 2, \dots, p$, then each factor $(G_i - B)$ can be developed as a convergent geometric series, resulting in:

$$y_t = \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \psi_3 \varepsilon_{t-3} + \dots \quad (4.16)$$

with the weights $\psi_i \rightarrow 0$ for $i \rightarrow \infty$.

This equivalence legitimizes the use of ARMA models, which are more effective than models with only the MA component. The process ARMA (p, q) is invertible if and only if the roots of the characteristic polynomial afferent to the moving average component $\Theta_q(B)$ are, in modulus, smaller than one. If for invertibility the operations performed for stationarity are repeated, it will be possible to see that an invertible ARMA (p, q) process is equivalent to an $\text{AR}(\infty)$ process.

Consequently, autoregressive or moving average processes comprising a large number of terms can be represented by AR-MA processes. This representation tends to have fewer parameters than the equivalent representations of the two pure processes.

4.5.2 Autoregressive Integrated Moving Average (ARIMA)

A non-stationary process on the average, as mentioned earlier, can be made in this manner when the evolutionary aspect can be traced to a polynomial of a certain degree. If y_t lacks stationarity then through a process of differentiation $\Delta = (1 - B)^d y_t$ it is possible to make it stationary.

The processes undergoing differentiation are said to be “integrated,” and the model used will be an ARIMA (p, d, q) . From a theoretical point of view, ARIMA processes exhibit infinite variance from which an undefined auto-correlation function is derived. In addition, the presence of the trend is reflected in the autocorrelation calculated on a sample basis, showing a decreasing trend that smooths out with a slow pattern. Similarly, the partial auto-correlation function tends to remain close to one for several lags.

The practice of “detrending” using the difference operator is based on the assumption that the trend-cycle is deterministic in nature and therefore that none of its erratic components affect the ARMA

part that remains after differences.

Therefore, we can say in conclusion that an ARIMA process is quite comparable to an ARMA process but taking into account the need to differentiate the series in order to make it stationary.

4.6 Time series modeling

The aim of this section is to illustrate all the steps involved in making the forecasts.

First of all, in order to forecast the amount of wind energy produced in the next 10 years, it was necessary to predict the factor required for production. Consequently, the forecasts are focused on the wind speed, and through the use of the power curve previously described and used it was possible to derive the wind energy produced.

4.6.1 Identification of potential explanatory variables

This section examines any relationships between wind speed and other explanatory variables.

This appears necessary for a twofold set of reasons:

- Identify any relationships with other variables in order to model the variable of interest (in the case under analysis, wind speed) through the use of multivariate ARMA/ARIMA models
- To be able to analyze the behavior of the variable of interest by means of a scenario analysis, i.e., an analysis in which certain conditions are imposed (in the case under analysis these conditions correspond to specific climatic configurations).

For the reasons listed above therefore, historical data inherent in:

- The wind speed at the site-specific height
- The temperature at the site-specific height
- The rainfall in the site area
- The humidity in the site area

For all variables listed above, historical data were used with hourly frequency from 2014-01-01 to 2024-01-01. In order to understand the potential existence of the above relationships, it was necessary to use the method of Ordinary Least Squares (which we will refer to as OLS). This model is used to test the relationship between a dependent variable (in the case under analysis, wind speed) and one or more independent variables. The mathematical formulation is the following:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon \quad (4.17)$$

Where:

- Y is the dependent variable
- $X_1, X_2, \dots, X_k$ are the independent variables

- $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are the regression coefficients representing the effect of the independent variables on Y ,
- ε corresponds to the random error representing the variation not explained by the model.

The aim of the OLS method is to identify the optimum value of the coefficients $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ by minimizing the sum of squares of the residual errors (ε). The table below represents one of several outputs obtained by exploring different configurations and different values of lags of the independent variables.

Model 1: OLS, using the observation 2014-01-01–2024-01-01 ($T = 87673$)

Dependent variable: Wind Speed

	Coefficient	Std. Error	t-ratio	p-value
const	30.4606	2.60297	11.70	0.0000
Temperature	0.0636334	0.0934801	1.901	0.0574
Temperature_1	−0.0522360	0.0815596	−1.655	0.0980
Temperature_2	0.00341644	0.0999245	0.1105	0.0912
Temperature_3	−0.105342	0.1282219	−0.733	0.1874
Mean dependent var	4.043974	S.D. dependent var		2.189520
Sum squared resid	16220.95	S.E. of regression		2.109548
R-squared	0.042733	Adjusted R-squared		0.043716
$F(4, 3645)$	28.32148	P-value(F)		3.33e−23
Log-likelihood	−7901.253	Akaike criterion		15812.51
Schwarz criterion	15843.52	Hannan–Quinn		15823.55
rho	0.506457	Durbin–Watson		0.986807

Analyzing the table in order, it can be seen that:

- The coefficient column represents the estimated values for each independent variable in the model
 - Regarding the temperature without delay this has a value of 0.063634 which indicates that, on average, a unit increase in temperature will result in a concomitant 0.063634 increase in wind speed
 - Temperature_1 corresponds to the temperature observed with one unit delay (in the case under analysis corresponding to one hour). This coefficient has a negative sign indicating that, on average, a one-unit increase in temperature at that time will result in a decrease in wind speed of 0.0522360
 - Temperature_2 corresponds to the temperature observed with two delays (i.e., two hours earlier). This coefficient appears to have a positive sign, so, as with what was explained in the case of the temperature observed at the same time, a unit increase in temperature at time $t - 2$ will result in a 0.00341644 increase in wind speed
 - Temperature_3 corresponds to the observed temperature at time $t - 3$ and, as in the case of the observed temperature at $t - 1$ there is also a negative coefficient, indicating that for

a unit increase in temperature at time $t - 3$ there will be, on average, a decrease in wind speed at time t of 0.105342

- The standard error represents a measure of how much the coefficients may vary around the mean value, and in the case under analysis these appear to be equal to:
 - Temperature: regarding this parameter it is indicated that its coefficient could vary around this value by about 0.0934801 units
 - Temperature_1: regarding this detection it is indicated that the actual value of the coefficient may vary around its actual value of 0.0815596 units
 - Temperature_2: regarding the temperature recorded at time $t - 2$ the value obtained is 0.0999245. This indicates that the calculated value of the coefficient may differ from its actual value by about 0.0999245
 - Temperature_3: finally, the value found at time $t - 3$ equal to 0.1282219, as seen for the previous values, indicates that the estimated coefficient may vary from its actual value by 0.1282219 units
- The t-ratio is a measure that provides preliminary information about the significance of each coefficient estimated in the model. In the case under analysis this appears to be equal to:
 - Temperature: about the detection of temperature at time t the t-ratio value (equal to 1.901) indicates that the variable has a significant deviation from the expected mean
 - Temperature_1: regarding the temperature observed with only one lag, instead, the t-ratio appears to be -1.655 which suggests, again, a significant deviation from the mean
 - Temperature_2: the t-ratio value of the observed temperature at time $t - 2$ is equal to 0.1105 and being very close to zero suggests that the independent variable has no significant effect on the wind speed (corresponding to the independent variable)
 - Temperature_3: finally, the t-ratio value for temperature at time $t - 3$ appears to be strongly negative and equal to -0.733 which, as explained above, suggests a potential low significance
- The p-value is a key statistical measure that, by testing the null hypothesis (defined as H_0) that assumes that there is no relationship between the variables in the model, provides insight regarding the significance of the independent variables with respect to the dependent variable. It must be made clear that for a result to be considered significant, it must have a p-value below a certain threshold. In the case under analysis, and as commonly chosen in similar analyses, a significance threshold of 0.05 was used, which corresponds to a 5% probability of obtaining the observed data if the null hypothesis is true.
The obtained p-values thus appear to be:
 - Temperature: the p-value associated with this detection is 0.0574, and therefore slightly above the limit value of 0.05, indicating that this parameter is not significant
 - Temperature_1: the p-value is 0.098, therefore well above the significance threshold, again indicating the non-significance of the observation

- Temperature_2: in this case, it is possible to note a slight decrease in the p-value compared to the value observed at time $t - 1$, but even in this case, the value does not appear to be acceptable at a significance level of 5%
- Temperature_3: the associated p-value (equal to 0.1874) for the detection at time $t - 3$ appears to be the highest of all, showing a decreasing trend of significance as the observed delay increases

The same analysis was carried out on all the potential variables mentioned above using even more than one of them at the same time and considering different delay configurations but the results obtained were still a non-significance of the variables in relation to wind speed.

Consequently, this implies that the ARMA/ARIMA model to be used will have to take into account only wind speed as a predictor of itself (i.e. a univariate model) and that, at least through the use of all the variables explored in this study, it is not possible to conduct a significant and accurate scenario analysis.

4.6.2 Identification and management of the seasonal component of the time series

The first step was to download historical data, with hourly frequency, on wind speed at the height considered and organize them in the format suitable for analysis. It appears necessary to specify that in order for the analysis to be mathematically possible it was necessary, as we will see later, to perform a manipulation of the time series, and in order to prevent the loss of information contained in the data, it was chosen to use historical hourly wind speed from 1940-01-01. At this point, since, as seen above, it was necessary to analyze the time series by going to consider its different components, amounting to:

- Trend
- Seasonality
- With the noises

This required running the Seasonal and Trend Decomposition using Loess algorithm, where Loess is a method for estimating nonlinear relationships. Through this operation it was possible to obtain a graphical display, shown below, of all 3 components for the entire historical series.

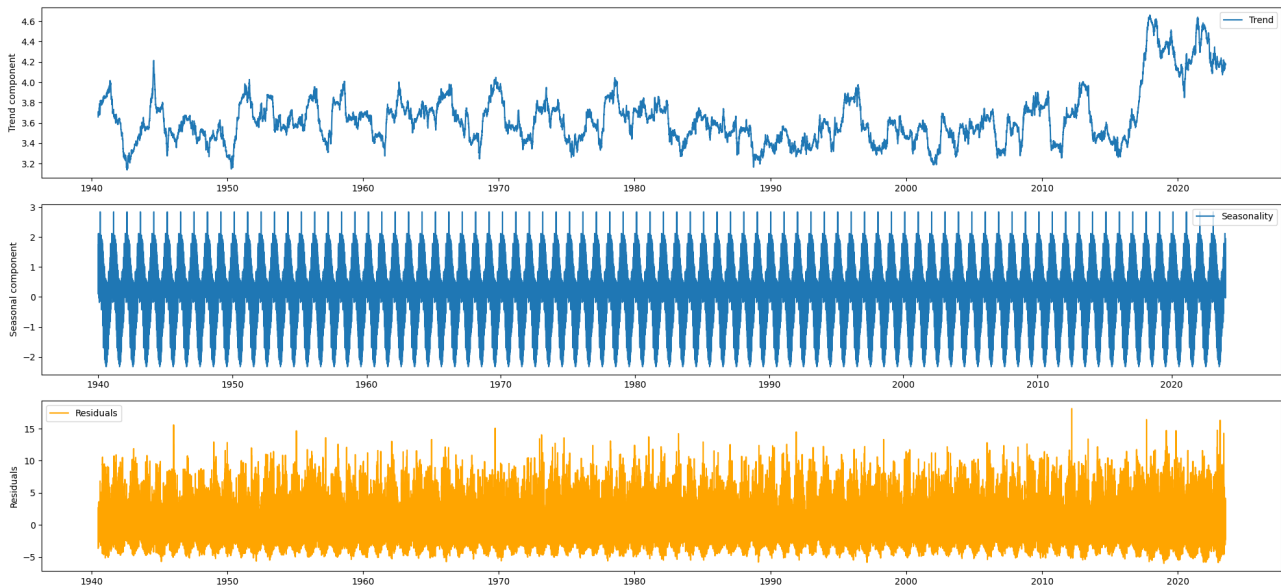


Figure 16: Seasonal and Trend decomposition using Loess)

From analysis of the graph it is immediately clear that:

- The trend appears to have been quite constant until beyond 2010, showing an oscillatory movement with variations of about 1 m/s
- The seasonal component, as plausible to expect, given the nature of the phenomenon and the analysis previously conducted when calculating historical production, appears to be strongly marked and to have a cyclical pattern with same frequency.
- The residual component, as specified several times already, corresponds to the remaining part of variation that is not explained by either trend or seasonality. This is precisely why it is necessary to make sure that these have mean close to 0 and that they distribute approximately as a Normal.

Regarding residuals, for the reasons mentioned above, it was necessary to verify their mean and distribution before proceeding with data modeling and analysis, obtaining the following results:

- For the average residual this appears to be 0.000867, which is a value close enough to 0, to be considered acceptable and thus in line with what is needed for further analysis.

This value of the mean appears necessary since it ensures that the model correctly captures all variations in the data as well as limits parameter bias, i.e., biased forecasts.

- The probability distribution, depicted in the figure below, appears to be similar to a normal one but with a longer right tail; this is due to the presence of outliers.

In any case, this representation appears to be adequate to satisfy the residual requirements of the stochastic models mentioned above, as these appear to be robust enough to perform adequately, even with moderate deviations from the normality of the residuals (especially when the sample is sufficiently large, as in the case under analysis). Moreover, even if the residuals are not exactly normally distributed, in many cases these can be approximated to normality especially in large enough samples where the central limit theorem is applicable.

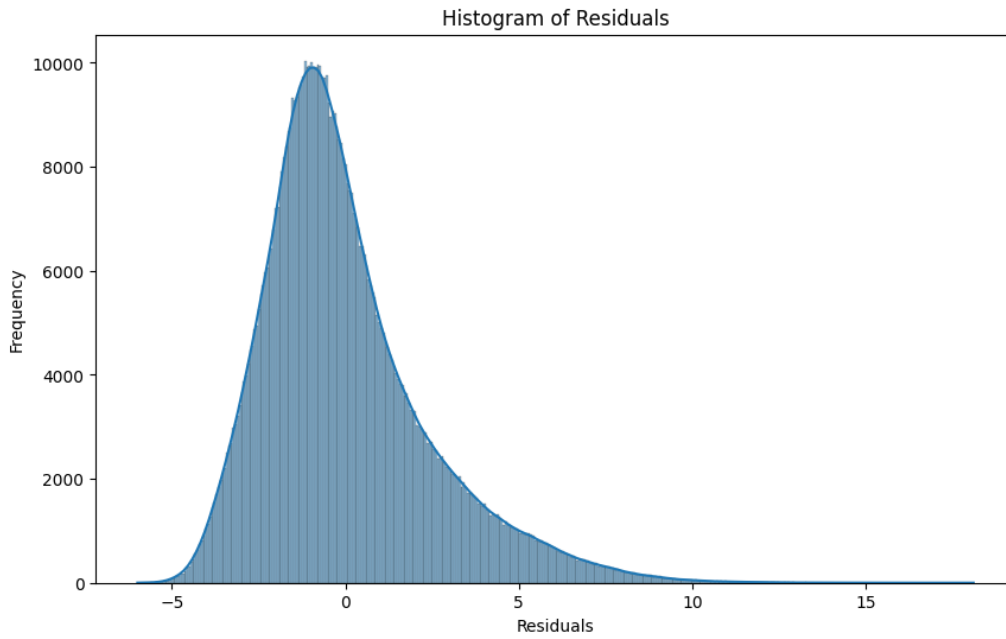


Figure 17: Residual component distribution

At this point, after having verified the goodness of the decomposition by means of residue analysis, in order to be able to run the forecasts on the timeseries, it was necessary to seasonally deseasonalise the entire dataset.

Deseasonalisation means removing the effect of seasonality altogether or reducing it. This process is crucial because it allows non-seasonal variations in the data to be isolated, enabling stochastic models such as ARMA/ARIMA to focus on irregular aperiodic fluctuations. In order to eliminate seasonality from the entire dataset, we relied on the results of Seasonal and Trend Decomposition using Loess (STL), which showed that this component was a constant function over time, thus suggesting a situation of additive seasonality. Consequently, by performing the difference between the original series and the seasonal component obtained, it was possible to obtain the hourly time series deseasonalised. The mathematical formula used is therefore the following one:

$$y_t = x_t - s_t \quad (4.18)$$

Where:

- y_t represents the time series deseasonalized
- x_t represents the observed value of the original time series
- s_t represents the seasonal component of the time series

In the figure below, the time series of historical wind speed adjusted by the seasonal component can be observed.

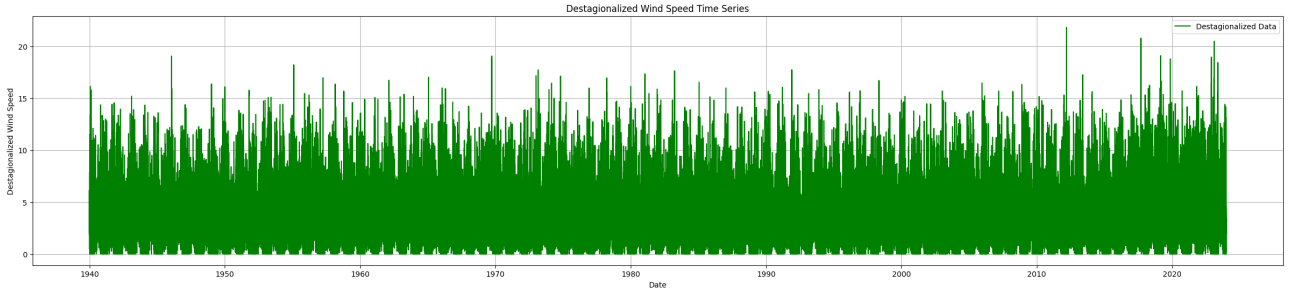


Figure 18: Wind speed deseasonalized time series

4.6.3 Data manipulation

Before continuing with the next step, it is necessary to make a specification. When long-term forecasting is attempted using an ARMA or ARIMA model, it is possible to observe the phenomenon of the decay of the AR (Autoregressive) and MA (Moving Average) terms, and this appears to be understandable from the general formulation of the two components of the model itself.

The general formulation of the Autoregressive (AR) model appears to be equal to:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t \quad (4.19)$$

Where:

- y_t : corresponds to the value of the time series at time t
- $\phi_1, \phi_2, \dots, \phi_p$: correspond to the Autoregressive coefficients, which are the terms that quantify the linear relationship between the current value of the time series and its past values.
- $y_{t-1}, y_{t-2}, \dots, y_{t-p}$: correspond to the past values of the time series
- ε_t : corresponds to the white noises (whose meaning was clarified earlier) at time t

As for a Moving Average (MA) model, the general formulation appears to be equal to:

$$y_t = \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} \quad (4.20)$$

Where:

- y_t : corresponds to the value of the time series at time t
- $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-q}$: it is the error term at time t : this term represents the random or irregular component of the series at the specified time t
- $\theta_1, \theta_2, \dots, \theta_q$: correspond to the coefficients of the model. Each coefficient θ_i represents the effect of the error term at time $t - i$ on the present value of the time series at time t

It can be seen that in both formulas the term y_t appears to be the sum of a coefficient (equal to ϕ regarding Autoregressive processes, and equal to θ , regarding Moving Average processes) multiplied by past observations and an error term.

Since these coefficients appear to be less than 1, by iterating the process in order to compute the forecasts it occurs an exponential decay of both coefficients leading the timeseries to converge to a single value, equal to the term ε (Messner *et al.*, 2013).

As it was mentioned, the decay of the influence of the AR and MA terms follows an exponential trend for each forward step that is predicted. Therefore, performing hourly forecasts for the next 10 years would mean having to calculate about 87600 steps, which would lead to an invalid result, as it is equal only to the ε component.

Consequently, in order to avoid the total decay of the terms, the time series was aggregated on an annual basis. Having already performed seasonal adjustment of the series, this aggregation also appears to be seasonally adjusted. Moreover, having the time series since 1940 will make possible to develop the analysis anyway on a dataset of adequate size to ensure the validity of the ARMA/ARIMA models.

4.6.4 Stationarity test

As discussed earlier, the initial choice regarding the need to perform differentiation is based on the stationarity or non-stationarity of the time series on which the analysis will be performed.

This feature can, sometimes, be identified by a graphical analysis of the time series and its sample autocorrelation function, but in order to obtain a more accurate and detailed quantitative assessment, it is possible to perform formal tests for unit roots on the autoregressive coefficients of the model. Early contributions to this type of test correspond mainly to the work of Dickey and Fuller, who proposed a test based on the conditional least squares estimator for an autoregressive process and the corresponding “t-statistic.”

In other words, to verify the stationarity of the time series the test tests the null hypothesis, corresponding to the presence of a unit root in the time series, and thus that the series is not stationary. The general formulation of the Augmented Dickey-Fuller (ADF) test formula corresponds to:

$$\Delta y_t = \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t \quad (4.21)$$

Where:

- Δy_t : corresponds to the first differentiation in the series.
- βt : is the term of deterministic seasonality
- y_{t-1} : corresponds to the value of time series lagged by a period
- γ : corresponds to the coefficient that tests the presence of the unit root
- δ_i : corresponds to the coefficients of lagged differences
- ε_t : corresponds to the white noises.

Since the time series under analysis was, as seen earlier, deseasonalized, the mathematical formulation in the present case appears to be equal to:

$$\Delta y_t = \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t \quad (4.22)$$

Where the meaning of the components is the same, as shown above.

Therefore, using the adapted formulation, the test will verify the following hypotheses:

- Null hypothesis H_0 : corresponds to the case where the time series has unit roots, so it will have a value of $\gamma = 0$ and, consequently, the series will not be stationary
- Alternative hypothesis H_1 : corresponds to the case where the time series has non-unit roots, so it will have a value of $\gamma < 0$ and, consequently, the series will be stationary.

The figure below shows the values obtained by using the ADF test.

ADF Statistic	p-value	1% Critical Value	5% Critical Value	10% Critical Value
-3.734839319068102	0.009827008807424436	-3.512738056978279	-2.8974898650628984	-2.585948732897085

Figure 19: Augmented Dickey–Fuller test results

These results clearly indicate that we can reject the null hypothesis and thus confirm the stationarity of the time series since:

- The value of the ADF statistic is -3.73, which suggests that the time series may be stationary
- The p-value obtained from the test is 0.0098 and, being below a default significance level of 0.05 suggests that the time series may be stationary
- Finally, the critical value seems to be -3.51, -2.897, and -2.586 for the 1% 5% and 10% significance levels, respectively.

Thus, the value of the ADF statistic was compared to these values and, being lower at all significance levels, suggests that the time series may be stationary.

Based on the results obtained, then it was possible to conclude that the series was stationary and therefore did not require any differentiation.

4.6.5 Autocorrelation Function (ACF)

After verifying the stationarity of the time series, and recognizing that no differentiation should be performed, at this point it appears necessary to identify the parameters p and q of the ARMA model. In order to identify the value of the parameter p , it is necessary to use the partial autocorrelation function (ACF).

This function is a statistical measure that evaluates the correlation between observations in a time series at different time intervals. In other words, the ACF shows how much past observations influence future observations in the time series.

The mathematical formulation of this function appears to be equal to:

$$\rho_y(k) = \frac{\text{Cov}(y_t, y_{t+k})}{\sqrt{\text{Var}(y_t) \cdot \text{Var}(y_{t+k})}} \quad (4.23)$$

Where:

- $\rho_y(k)$: represents the ACF at lag k

- y_t, y_{t-k} : are, respectively, the time series observations at times t and $t - k$
- $\text{Cov}(y_t, y_{t-k})$: is the covariance between the observations y_t and y_{t-k} , i.e., the average of the product of the deviations from the mean term
- $\text{Var}(y_{t+k})$: are, respectively, the variances of the observations y_t and y_{t+k} .

For the identification of significant lags for the model under analysis, we used a confidence interval of 95% since this interval allows:

- To control the risk of erroneously including delays in the model that are not effectively auto-correlated
- To identify significant lags with a probability of 95% thereby achieving a good degree of accuracy in the identification process.

Using the logic described above, then it was possible to identify the significance of the autoregressive parameters and, using, as mentioned, a confidence interval of 95%, it was possible to proceed with the graphical analysis and identification of significant lags.

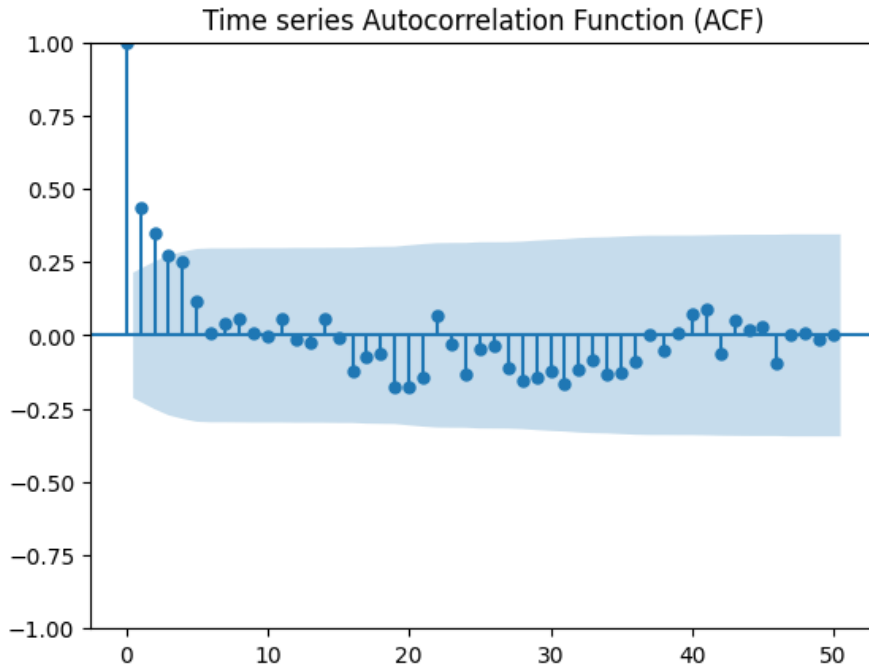


Figure 20: Time series Autocorrelation Function (ACF)

Through analysis of the figure, it is possible to see that the lags at times $p = 1$, $p = 2$ and $p = 3$ appear to be greater than the confidence interval and, consequently, it will be necessary, as it will be shown later, to identify the most explanatory one also in relation to the parameter q .

4.6.6 Partial Autocorrelation Function (PACF)

After estimating the possible orders of the parameter t for the ARMA model formulation to be complete, it also needs to identify the order of the parameter q . For this purpose, it is necessary to compute

the partial autocorrelation function (PACF), whose aim is to measure the correlation between observations at a given lag by taking into account only the direct correlation between observations while ignoring intermediate effects (contrary to what is done with ACF). The general mathematical formulation of such a function is the following:

$$\phi_k = \frac{\text{Cov}(y_t, y_{t-k} | y_{t-1}, y_{t-2}, \dots, y_{t-k+1})}{\sqrt{\text{Var}(y_t | y_{t-1}) \cdot \text{Var}(y_{t-k} | y_{t-1}, y_{t-2}, \dots, y_{t-k+1})}} \quad (4.24)$$

Where:

- ϕ_k : represents the value of PACF at lag k
- $\text{Cov}(y_t, y_{t-k} | y_{t-1}, y_{t-2}, \dots, y_{t-k+1})$: represents the covariance between observations y_t and y_{t-k} conditioned to all intermediate observations $y_{t-1}, y_{t-2}, \dots, y_{t-k+1}$.
In other words, it is the covariance between observations at a lag k after removing the effect of intermediate observations
- $\text{Var}(y_t | y_{t-1})$: corresponds to the conditional variance of y_t given y_{t-1}
- $\text{Var}(y_{t-k} | y_{t-1}, y_{t-2}, \dots, y_{t-k+1})$: corresponds to the conditional variance of y_{t-k} given $y_{t-1}, y_{t-2}, \dots, y_{t-k+1}$

Similarly to what was done previously for the identification of the correct order of p , again it was necessary to use a confidence interval of 95% to maintain criterion consistency and, in addition, the same advantages described above can be gained. At this point, an analysis of the PACF graph was performed to identify significant delays.

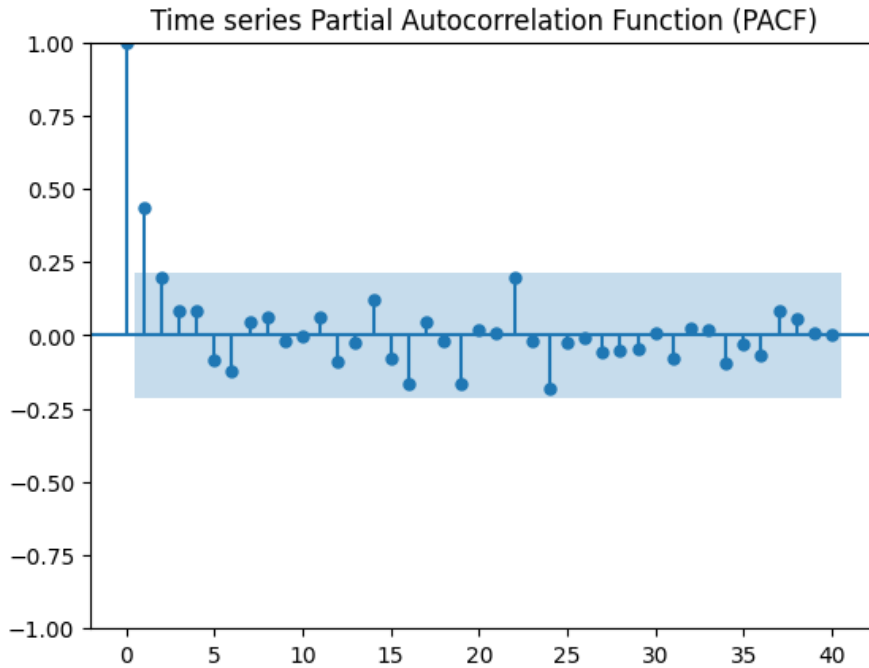


Figure 21: Time series Partial Autocorrelation Function (PACF)

The analysis of the above graph made possible to identify $q = 1$ and $q = 2$ as significant lags since these appear to be above the specified confidence interval.

4.6.7 Model identification

At this point, having run the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), and having different values for both p and q , it is necessary to identify which combination of these parameters appears to be the best. For this purpose, what are called the “Information Criteria” have been used.

The first information criterion used in the following analysis is the Akaike Information Criterion (AIC). This criterion is an estimator of the error generated by the prediction in the time series and is based on the concept that, when a model is used to represent the data, it is never perfectly accurate; consequently, the AIC estimates the amount of information lost for each possible model by expressing this loss as a numerical value.

Consequently, based on the AIC, it will be possible to identify the model that generates the lowest information loss.

Moreover, before continuing with the description of this criterion, it seems useful to point out that it also plays a fundamental role in preventing the risk of overfitting, i.e. the risk of the model adapting to the data not because of its goodness but because of the excessive number of parameters in relation to the data. The mathematical formulation of this model is the following:

$$AIC = -2 \cdot \ln(L) + 2k \quad (4.25)$$

Where:

- L : is the parameter corresponding to the maximum likelihood function, representing the probability that the model generates the observed data
- k : is the number of parameters in the model.

The logarithm is applied to L since this allows numerical stabilization to be obtained: indeed, sometimes the parameters of L can be very small, and applying the logarithm reduces the risk of numerical errors due to the size of the values of L . On the other hand, the parameter k is multiplied by 2, which is functional to control the complexity of the model. In fact, increasing the number of parameters increases the complexity of the model, and therefore the ‘penalty’ is also increased.

Before proceeding to analyze the results obtained using of the AIC, the second Information Criterion used will be analyzed.

This choice was made since, sometimes, the use of a single criterion leads to the identification of an incorrect parameter order and, to achieve a greater accuracy, it was decided to use it together with a second one.

The second criterion used is the Bayesian Information Criterion (BIC), a metric used for model selection and developed on the basis of Bayesian principles. Like the Akaike Information Criterion, this also attempts to balance the accuracy of model fit with the complexity of the model itself but, differently from the previously analyzed criterion (and as will also be observable in the mathematical representation), this criterion also applies a penalty based on the number of observations used. Consequently, a greater number of observations leads to a greater penalty for more complex models. The mathematical formulation appears to be equal to:

$$BIC = -2 \cdot \ln(L) + k \cdot \ln(n) \quad (4.26)$$

The parameters used are the same as those described in the AIC formula, but, in contrast to what we have seen before, this formulation includes $\ln(n)$ which indicates precisely a logarithmic growth of the penalty as a function of the number of observations included.

At this point, having calculated both metrics, the table below shows the results obtained.

Values of AIC and BIC for parameter combinations

p	q	AIC	BIC
1.0	1.0	1526.9304514797777	1536.653718675151
1.0	2.0	1528.932294294079	1541.0863782882955
2.0	1.0	1528.9308994983187	1541.0849834925352
2.0	2.0	1528.1755274461236	1542.7604282391835
3.0	1.0	1530.69927001776	1545.28417081082
3.0	2.0	1532.5371709918795	1549.5528885837828

Figure 22: Results of the Akaike Information Criterion and Bayesian Information Criterion

In the above table it can be seen the results of both criteria for all six possible combinations of p and q . As also appears clear, the optimal combination of parameters is $p = 1$ and $q = 1$, since for both BIC and AIC the minimum values (thus representing the least information loss) occur in that combination.

Thus, in light of this analysis, it is possible to state that the time series will have to be analyzed using an ARMA(1, 1) model.

4.6.8 Model fitting and forecasting

At this point, after identifying the correct model formulation, the specific mathematical formulation that will be used for the time series analysis appears to be as follows:

$$y_t = c + \phi y_{t-1} + \theta \varepsilon_{t-1} + \varepsilon_t \quad (4.27)$$

Where:

- c : corresponds to a constant term
- y_t : corresponds to the value observed at time t
- ϕ : corresponds to the coefficient of the autoregressive (AR) component of the formula
- y_{t-1} : corresponds to the value observed at time y_{t-1}
- θ : is the coefficient of the moving average (MA) component of the formula
- ε_{t-1} : is the random of the error

As can be seen, the formulation appears to have the same form as the general one shown above but, given the order of the model, the number of observed lags is one. The results of the model fit are summarized in the table below.

Dep. Variable:	Destagionalized_Wind_Speed	No. Observations:	84			
Model:	ARIMA(1, 0, 1)	Log Likelihood	-759.465			
Date:	Thu, 06 Jun 2024	AIC	1526.930			
Time:	01:05:31	BIC	1536.654			
Sample:	12-31-1940	HQIC	1530.839			
	- 12-31-2023					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	3.199e+04	923.351	34.643	0.000	3.02e+04	3.38e+04
ar.L1	0.8826	0.107	8.267	0.000	0.673	1.092
ma.L1	-0.5609	0.146	-3.835	0.000	-0.848	-0.274
sigma2	4.123e+06	6.47e+05	6.375	0.000	2.86e+06	5.39e+06
=====						
Ljung-Box (L1) (Q):		0.00	Jarque-Bera (JB):		6.64	
Prob(Q):		0.99	Prob(JB):		0.04	
Heteroskedasticity (H):		2.24	Skew:		0.65	
Prob(H) (two-sided):		0.04	Kurtosis:		3.47	
=====						

Figure 23: ARMA model fitting results

From the above table it can be understood that:

- Log. Likelihood: this indicator expresses the maximum likelihood of observing the accurate data with the specified model. In the case under analysis this value is equal to -759.46, which indicates that there is a large component of error which, however, appears to be understandable since, as clarified at the beginning of the study, the precision of the estimates decreases as the time horizon one attempts to estimate, increases and, in this case, appears to be equal to 10 years
- Akaike Information Criterion (AIC): such an indicator, as explained above, is an indicator that specifies which order of p and q is better to choose
- Bayesian Information Criterion (BIC): this metric, seen above, plays the same role as AIC, has the role of reinforcing the evidence, concerning the order of the model to be chosen, obtained through the calculation of AIC
- Hannan-Quinn Information Criterion (HQIC): This is also an information criterion used for model selection. This metric appears to be similar to AIC and BIC, but provides even greater penalties than both as the number of parameters increases.

The mathematical formulation appears to be equal to:

$$HQIC = -2 \cdot \ln(L) + k \cdot \ln(\ln(n)) \quad (4.28)$$

- Covariance Type (opg): this information specifies that the covariance of the model coefficients was calculated using the Outer Product of Gradients (OPG) method.
This method for estimating the parameter covariance matrix is designed to provide robust parameter estimates even in the presence of deviations from normality or other, minor, violations of the assumptions of the model's assumptions

- const: represents the constant term of the model
- ar.L1: represents the autoregressive coefficient (ϕ) of order one
- ma.L1: represents the moving average coefficient (θ) of order one
- sigma2: represents the residual variance of the model
- coeff: this column indicates the estimated coefficients for each variable in the model, which, moreover, are equal to:

- As for the constant term this has a value of 3,199e+04 which indicates that when all other variables in the model are zero, this term is expected to have an average value of about 31,990.

This appears consistent with the aggregation performed in the “Data Manipulation” section in which the data were rearranged on an annual basis for the reasons previously described

- The autoregressive coefficient (ar.L1) has an estimated value of about 0.88, suggesting the existence of a strong linear dependence of wind speed on its past values
 - The coefficient of moving average (ma.L1) has an estimated value of about -0.56, indicating a negative and significant dependence of the time series on past errors of the model
- Std err: This column represents the standard error associated with each coefficient in the model. This measure indicates the precision with which each coefficient is estimated. Thus, the results of the model are equal to:

- The standard error of the constant appears to be 923,351 which indicates that, under the condition where all variables in the model are 0, the predicted value may vary by about 923,351 units around the estimated value of the constant term
- As for the ar.L1 coefficient this has a standard error of about 0.107 indicating very good precision obtained in estimating the coefficient of the autoregressive part
- As for the ma.L1 coefficient this has a standard error of 0.146, indicating, as already seen for the case of the ar.L1 coefficient, very good precision in estimating the value of this coefficient

- The z column in the table represents the value of the “z-statistic” test for each coefficient estimated by the model.

This value is calculated as the ratio of the estimated coefficient to its standard error and is used to test the null hypothesis that the true value of the coefficient is zero (and that, consequently that the coefficient is not significant). The mathematical formulation of such a test is equal to:

$$z = \frac{\text{Coefficient}}{\text{Standard error}} \quad (4.29)$$

The values obtained for this metric are as follows:

- As for the constant term, it has a very high value indicating that this term is very significant in the model and that consequently the probability of it being 0 is extremely low (as will be observed later by analyzing the next column)
- The coefficient ar.L1 also has a very high value indicating that this coefficient is highly significant and with very low probability is zero
- The value of the moving average coefficient is approximately at -3.85 which, although negative, appears elated in absolute value indicating, again, that the first-order MA coefficient is significant.

In general, regarding the “z” column, it is possible to say that all values, regardless of sign, are very high, indicating that each is highly significant and contributes substantially to the model.

- Column $P > |z|$ shows the p-values associated with the hypothesis tests for each estimated coefficient in the model.

In other words, the p-value indicates the probability of obtaining a value of the test statistic (z) so extreme that we must accept the null hypothesis that the coefficient is zero.

The values obtained for the different components of the model are equal to:

- the p-value for the constant term is 0.000 indicating that the probability of the coefficient being zero is extremely low and, consequently, it is possible to state that the constant term is highly significant and cannot be ignored
- A p-value of 0.000 can also be observed for the coefficient ar.L1 indicating, similarly to the previous case, the rejection of the null hypothesis and the establishment of the significance of this coefficient
- Finally, it was seen for the previous two parameters, the coefficient ma.L1 also appears to be equal to 0.000, and therefore highly significant
- The columns “0.025” and “0.975” indicate the external limits of the 95 percent confidence interval for each of the estimated coefficients.

This confidence interval indicates the range of values within which the true value of coefficient is expected to be found 95% of the time.

The results obtained for the parameters under analysis are as follows:

- The constant term has a confidence interval between 30200 and 33800 which, of course, does not include zero, confirming that the constant term is significantly different from 0.
- The same argument made for the constant term can be applied regarding the coefficient ar.L1 equal to 0.673 and 1.092. This range again confirms that the value of the coefficient cannot take the value of zero, and it is significant
- Also in the case of the coefficient ma.L1, it can be observed that this is defined in an interval and equal to the range -0.848 and -0.274, confirming again, the impossibility of the coefficient to be zero and the significance
- Ljung-Box (L1) (Q): this test checks the null hypothesis that the first k autocorrelations of the residuals are all zero, so essentially six residuals of the model are independent and uncorrelated. The obtained result of 0.00 indicates the presence of no significant correlation in the residuals

- Prob (Q): this measure shows a value of 0.99, indicating that the residuals do not show significant correlation and therefore the model has adequately captured the structure of the data
- Jarque-Bera (JB) is a test used to check the null hypothesis that the residuals follow a normal distribution by considering the skewness (i.e., asymmetry) and kurtosis (i.e., flattening) values of the distribution of the residuals.
A value of 6.64 suggests that there may be a deviation of the expected values for a normal distribution
- Prob (JB): shows the possibility of rejecting or accepting the null hypothesis (at a significance level of 5% that the residuals follow a normal distribution. In the case under analysis, even the value shown by this metric indicates that the residuals may not distribute as a Normal. This result appears to be in line with the slight deviations noted during the analysis of the series residuals and may be justified by the sample size and the presence of outliers.
- Heteroskedasticity Test (H): this test checks the null hypothesis that the residuals have constant variance (homoschedasticity). The value shown is 2.24, which indicates that the variance may not be constant
- Prob (H) (two-sided): the value shown by that measure of 0.04 appears to be below the threshold of 0.05 and thus suggests the presence of heteroschedasticity.
This aspect may require some reworking but, considering the presence of outliers in the time series, the length of the estimated time horizon and the metrics shown by the other indicators, this metric can still be considered acceptable and compatible in ensuring an adequate level of accuracy of the model
- Skweness: this statistical measure, as also explained above, shows the symmetry of the distribution in relation to a Normal distribution.
In the case under analysis the value obtained is 0.65 showing a slight positive skewness which, in other words, implies that the residuals tend to have, on the right side of the distribution, more extreme values
- Kurtosis: This indicator shows the flattening of the distribution of residuals from a normal distribution (whose kurtosis is 3).
The value obtained is 3.47, indicating that the distribution of residuals has slightly “heavier” tails than a normal one, which could indicate the presence of outliers

The forecasts obtained through the "step ahead" estimation process are as follows:

Predicted_Value	
Date	
2024-12-31	35614.971902
2025-12-31	35189.155653
2026-12-31	34813.325785
2027-12-31	34481.614420
2028-12-31	34188.842505
2029-12-31	33930.438957
2030-12-31	33702.369287
2031-12-31	33501.072616
2032-12-31	33323.406072
2033-12-31	33166.595725

Figure 24: Forecast (10 years)

The values shown correspond to the sum of hourly wind per year for the next 10 years.

As can be seen, the values appear decreasing and this is due, as it was illustrated above, to the decay of the autoregressive (AR) and moving average (MA) parameters.

At this point, in order to be able to use the data according to the original frequency, it will be necessary to bring them back to the original (hourly) frequency through what is called a downscaling process.

4.6.9 Downscaling Process

As shown by the table in the previous paragraph the forecasts were calculated as the annual sum of hourly wind speed but, in order for the analysis to be consistent with the data originally used to estimate wind production it was necessary to render them with hourly and, for this purpose a frequency downscaling process was used. Frequency downscaling is a method used to transform aggregate data on larger time scales into estimates on shorter time scales. This process turns out to be fundamental and widely applied in several areas such as engineering and meteorology, where, for forecasting reasons, it is necessary to have data with more granular time resolution. Downscaling processes are divided into two main types, amounting to:

- Dynamic downscaling: a type of process that, through the use of high-resolution physical models simulates processes on shorter time scales
- Statistical downscaling: this type uses statistical relationships between observed historical data to change the frequency

A statistical downscaling based on incidence rates was used in the following analysis.

This approach uses the historical distributions of variables on short scales to redistribute the data represented on long scales to shorter scales. In addition, in order to obtain a higher precision of the incidence of each hour within the annual total, the incidence for each hour of each day for each year

was calculated. Then, it was calculated the average of the hourly incidences across all days to obtain the average hourly incidence. The formula used for this purpose is the following:

$$P_i = \frac{\sum_{i=1}^N \sum_{j=1}^M \sum_{k=1}^L V_{ijk}}{N \cdot M \cdot L} \quad (4.30)$$

Where:

- P_i : Represents the average incidence of each hour of each day of each year
- N : Corresponds to the number of years considered
- M : Represents the number of days in each year
- L : Corresponds to 24, which is the number of hours in each day
- V_{ijk} : Represents the incidence value for hour k of day j in year i

After performing this process, it will then be possible to know the value of the wind speed (expressed in m/s) for each hour of the next 10 years, and based on this data (which will be shown in the next section), combined with the power curve shown initially, it was possible to perform the estimates of wind production for each hour of the next 10 years.

4.6.10 Results

The following section will show the results of wind speed forecasts for the next 10 years as well as the related wind production forecasts and the main statistical measures functional for interpreting and comparing the results.

As shown in the following table, after the downscaling process it was possible to observe the predicted wind speeds for each hour, which are equal to:

Predicted_hourly_values	
date	
2024-01-01 00:00:00	2.352669
2024-01-01 01:00:00	2.439542
2024-01-01 02:00:00	2.384626
2024-01-01 03:00:00	2.536515
2024-01-01 04:00:00	2.342594
...	...
2033-12-31 19:00:00	1.692248
2033-12-31 20:00:00	1.417863
2033-12-31 21:00:00	2.190933
2033-12-31 22:00:00	2.778728
2033-12-31 23:00:00	3.559384

Figure 25: Hourly forecast (10 years)

In addition, as will be shown in the next figures, in order to have an overview of the predicted wind speed, it was decided to graphically represent the entire wind speed.

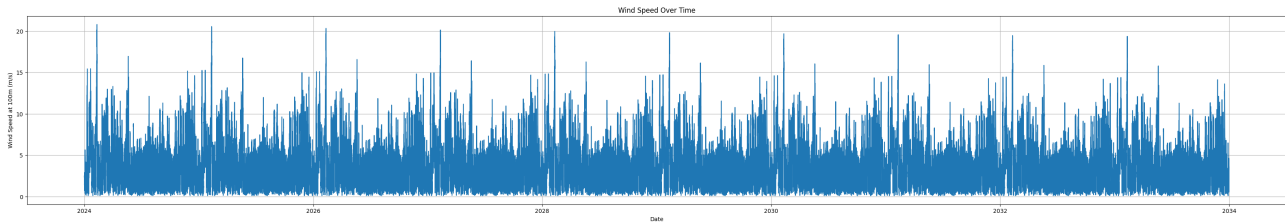


Figure 26: Hourly forecasted wind speed time series (10 years)

As shown by the figure there appears to be a cyclicity of the estimated values in the different years, and, in order to better understand the structure, the trends of the month's velocity for each month of each year have also been shown.

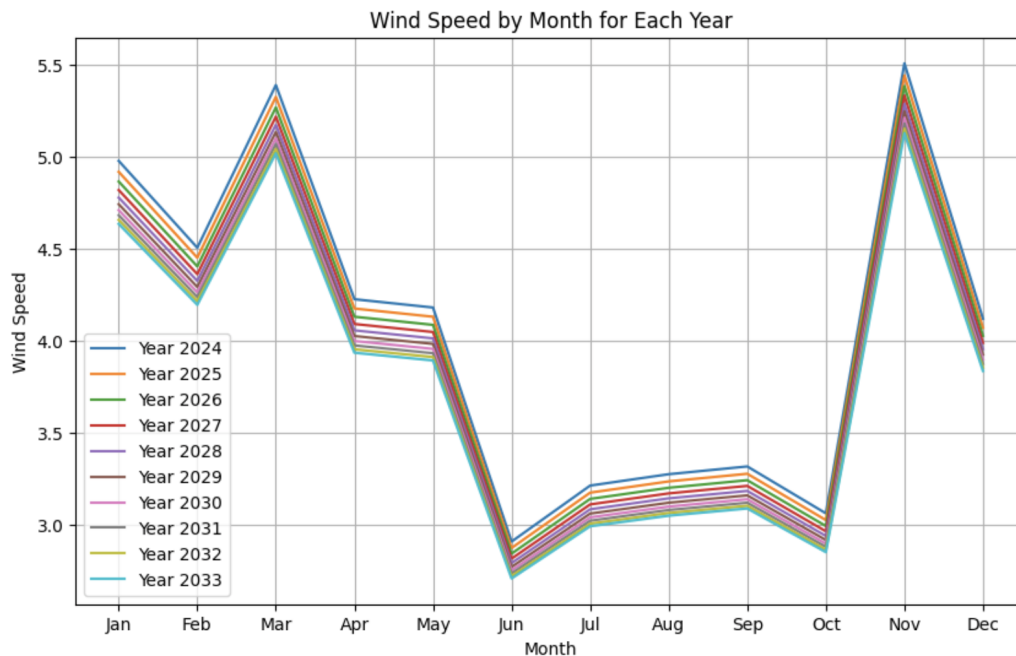


Figure 27: Predicted monthly patterns of wind speed (10 years)

What the images show appears to be in line with what we have seen previously in fact:

- From both the first and second graphs, it is possible to see that the wind always follows the same pattern, and this is due to the fact that, in performing the downscaling process, it was decided to use a single average value (for each hour of each day) representative of the percentage incidence
- From the second graph it is also possible to see that the patterns are always decreasing, and this is in line with the theory of Moving Average Autoregressive models, which always show a decreasing trend in the predicted values

After graphically analyzing the predicted wind patterns for the next 10 years, the predicted wind energy produced was calculated.

To obtain this forecast, it was used, as seen above, the technique of interpolation of the power curve in order to be able to calculate the production for any wind value in the range allowed by the power curve itself (range corresponding to the values between the minimum speed required to turn on the plant and the maximum speed beyond which the plant shutdown occurs). First of all, as shown by the figure below, the entire estimated time series was represented.

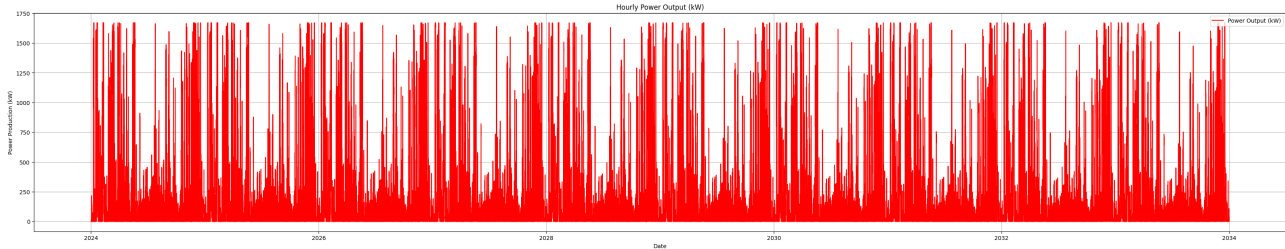


Figure 28: Time series of the expected hourly production (kW) (10 years)

As can be seen from the graph the projected wind production appears to touch its maximum values at about 1670 (value corresponding to the maximum of the power curve) kW and appears to take on a structure such that roughly at the beginning and middle of the year the maximum peaks of production are recorded. This last feature is not surprising since, as seen in the Betz formula, the density of the air appears to be a relevant factor in the calculation of production and, this density, appears (as shown in the formula) to be directly influenced by the temperature.

Subsequently, in order to be able to observe the temporal evolution of production, it was also decided to represent the annual production for the next 10 years.

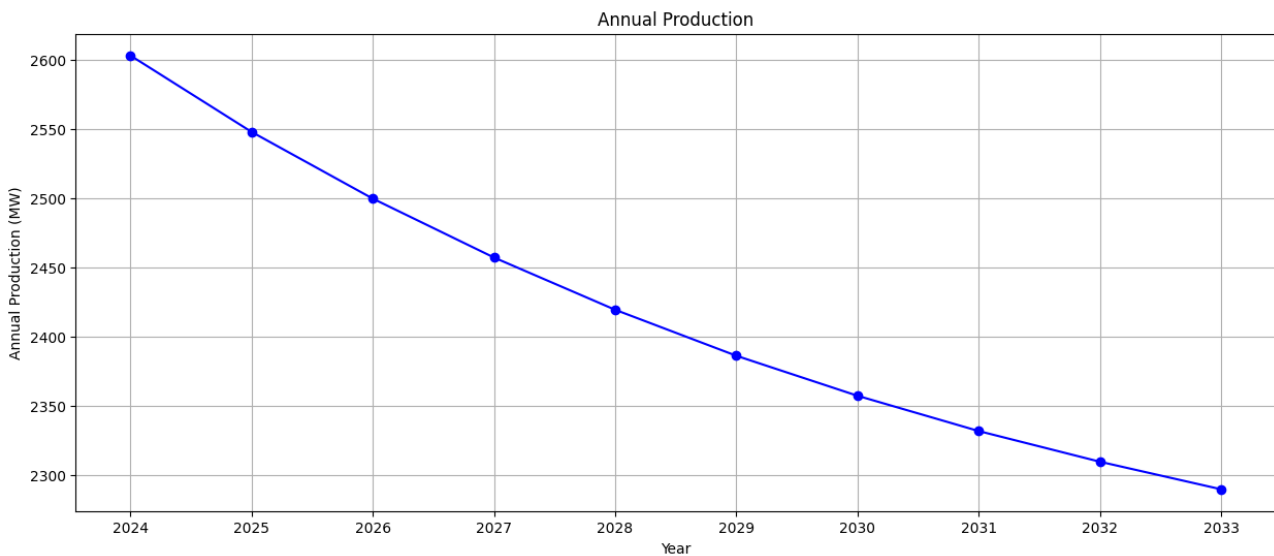


Figure 29: Estimated annual production trend (10 years)

As shown by the figure again, and even more clearly, it can be seen that the forecast, as expected, takes a downward trend. Moreover, it should be pointed out how the forecasts show annual production values not very different from those stated by the site owners, in fact, the average production forecast for the next 10 years appears to be 2425MW that is higher by 125MW.

Finally, before proceeding with the comparison of the models performed against historical values, it appears relevant (both for greater interpretation of the characteristics of the predicted time series of wind speed and to allow comparison with historical data) to observe the main statistical measures, which appear to be:

Mean	Variance	Standard Deviation	Kurtosis	Skewness
196.33598958441027	144169.34060902166	379.69637950475857	4.873604843026309	2.3693826560711093

Figure 30: Main statistics of the forecasted time series

From the analysis of the table, it can be understood that:

- The average hourly production is expected to be about 196kW
- The variance of the predicted values is 144169, this value is easily understood since when the wind speed appears to be lower than the on speed or higher than the off speed the production will be zero
- The standard deviation, which uses the same unit of measurement as the time series, indicates that the predicted values differ by 379kW from the mean value. Again, this value appears understandable since the standard deviation is equal to the square root of the variance
- The Kurtosis appears to be about 4.87 which indicates that the tails of the distribution are heavier than a Normal distribution
- The Skewness, equal to about 2.37, shows positive asymmetry and consequently a greater extension of the right tail of the distribution.

Chapter 5

Comparison

In this chapter, a comparison of the main evidence, graphical and statistical, of the obtained models will be carried out by making a comparison with what have been the values of past wind production. The comparison will mainly focus on a graphical comparison and on a comparison of statistical metrics in order to be able to understand how, the different models predicted the future wind path. First, it appears essential to analyze the time series graph of historical production in comparison with that obtained by applying the ARMA(1,1) model. In order to have a clearer interpretation of the graph, the two time series have been represented in one graph but marked by different colors.

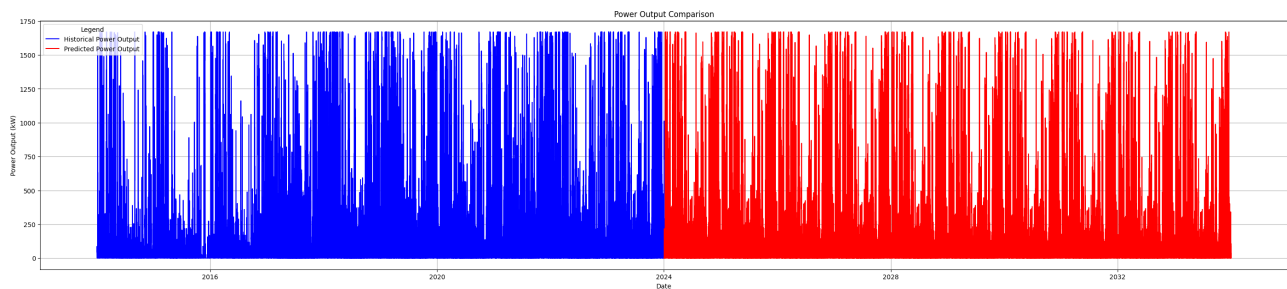


Figure 31: Historical and forecasted time series

From the graph, it is possible to see that the predicted data adequately fit the historical series by continuing the pattern as early as 2024. It is also possible to see that the predicted production peaks appear to be at the same times of the year as occurred in the historical data.

Observing the production peaks, it is possible to note how these, despite having a similar value to the historical one, are decreasing and this is due to the formulation of the ARIMA model undergoing the effect of coefficient decay and, with them, a consequent decrease in the estimated values.

Finally, the graph shows adequate accuracy of the model ensuring, once again, that the time horizon used for the calculation was adequate.

At this point, after graphically analyzing the two time series, it seemed necessary to also analyze, comparatively, the main statistical metrics in order to understand the extent of possible variations. The tables below represent, in order, the table concerning the main statistical measures of the original time series and, immediately following, that concerning the time series forecasted.

Mean	Variance	Standard Deviation	Kurtosis	Skewness
205.67268459558574	148839.7836963778	385.7975942076075	5.002556615965608	2.3895293450888855

Figure 32: Historical wind energy production: main statistical measures

Mean	Variance	Standard Deviation	Kurtosis	Skewness
196.33598958441027	144169.34060902166	379.69637950475857	4.873604843026309	2.3693826560711093

Figure 33: Main statistics of the forecasted time series

From the comparative analysis of the two tables regarding statistical measures, the main evidences are the following ones:

- Mean: this value, as illustrated throughout the analysis, represents the average hourly wind production value.

Looking at the tables, it can be seen that the average hourly production obtained through the model ARMA (1.1) is lower by about 9.3376kW per hour. This difference can be attributed to three main reasons:

- The decay of the Autoregressive (AR) and Moving Average (MA) coefficient causes, as explained, a decrease in the predicted value and consequently a decrease in the mean value
 - The model errors are a second aspect that needs to be considered since, having to predict a very large time horizon, there is a large uncertainty and, most likely, part of this decrease in the mean value is attributable to the error term
 - A final aspect to consider is unforeseen variations in historical data, that is, unpredictable variations in the data that could not be modeled (such as exceptional events or large-scale variations).
- Variance: This parameter corresponds to the dispersion of the data around the mean value of the distribution and appears to be lower by about 4670 in the estimated model, which indicates that the model was able to slightly reduce the variance from the original variance.
This variance can be explained by the following reasons:
 - The model may have successfully captured the main trends of the original time series. This could reduce variance since the model is able to "explain" a significant portion of the variation in the data by focusing on more predictable components of the time series
 - The parameters used in calculating the model were able to adjust for the variance in the data, thereby reducing the variance in the predictions
 - Standard deviation: As mentioned above, it is a measure that indicates the amount of deviation of observations from the mean value.

This value appears to be lower for reasons that can be traced back to what was expressed when the variance was discussed, and this is because the standard deviation is the square root of the variance

- **Kurtosis:** This measure indicates the "heaviness" of the tails of the distribution of the time series values, and as can be seen from the comparative analysis of the two tables, the kurtosis of the predicted time series appears to be slightly lower (by about 0.2).
This variation indicates that the shape of the distribution of forecast values takes a conformation characterized by a lower presence of extreme events
- **Skewness:** As mentioned this measure indicates the symmetry of the distribution, and as can be seen from the tables, the value of this measure of the predicted time series appears to be 0.02 lower than the same measure of the original time series.
This change is almost imperceptible and represents a slight decrease in the positive asymmetry. This variation is related to the reduction in outlier values (as moreover confirmed by the kurtosis seen above).

To conclude this first comparison, it is possible to state that the forecasts made appear to be similar to the historical data (thus confirming the meteorological theory of the persistence of phenomena) while still presenting slight variations due, as clarified during the analysis of the different metrics, to the very nature of the model used.

The second comparative analysis will be conducted between historical data and Monte Carlo forecasts. The two graphs below illustrate the patterns of historical and projected wind production

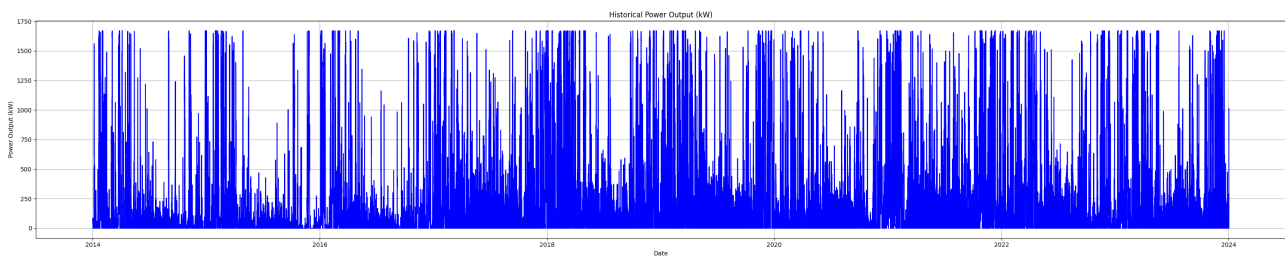


Figure 34: Historical wind power production

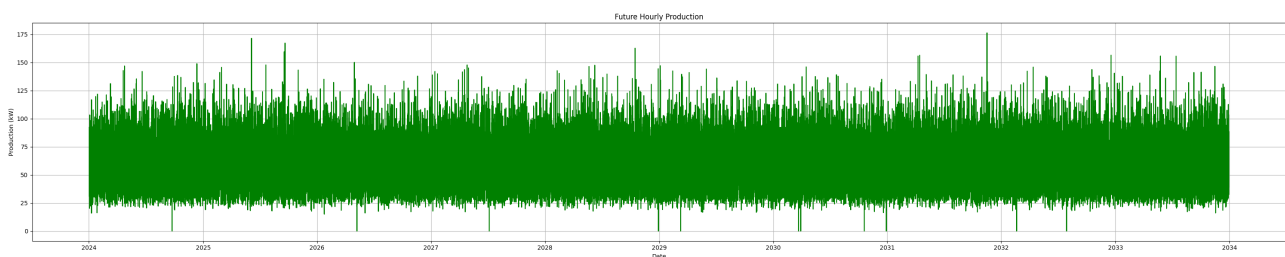


Figure 35: Wind production calculated by average Monte Carlo simulation (10 years ahead)

As is immediately clear, and understandable, through looking at the two graphs, the graph representing the expected output estimated by averaging the Monte Carlo paths appears to be highly concentrated around the mean and with little variability. This aspect is due to the path averaging operation that was performed, but such a representation would be too limiting, so it was chosen to compare historical production with that calculated using the median path of the Monte Carlo simulation. The representation of the entire time series, including both the historical and predicted time series is shown in the figure below.

The last figure represents more accurately and realistically a possible pattern of wind production over

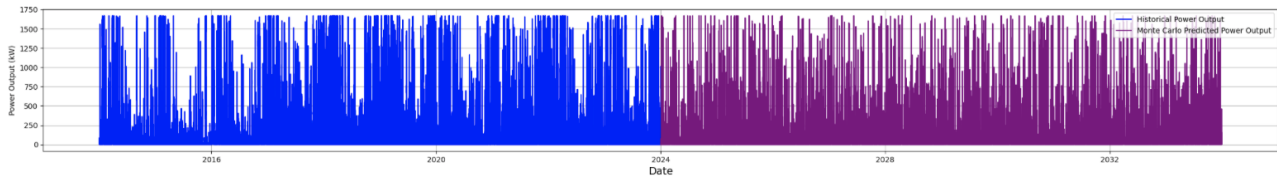


Figure 36: Historical and forecasted time series

the next 10 years.

Through observation of the graph, we can see that the peaks appear to be more distributed over the course of the year and not concentrated in specific months. The peak production is always around 1670 (a value representing the maximum amount that can be produced as shown by the power curve displayed at the beginning of the following discussion).

At this point, to better understand the differences between the time series production values and one of the productions calculated by one of the estimated Monte Carlo paths, it appears again necessary to analyze the tables representing the main statistical metrics.

Mean	Variance	Standard Deviation	Kurtosis	Skewness
205.67268459558574	148839.7836963778	385.7975942076075	5.002556615965608	2.3895293450888855

Figure 37: Historical wind energy production: main statistical measures

Mean	Variance	Standard Deviation	Kurtosis	Skewness
190.7237409971359	127672.8030836949	357.31331221169876	6.499735887207874	2.5976987422304703

Figure 38: Monte Carlo median path wind energy production: main statistical measures

- **Mean:** as observable, the value of the mean of the predicted time series appears to be 10kW lower for each hour.
This seems understandable since, in the above table, only one path is shown and this could be characterized by a lower average
- **Variance:** As mentioned earlier the value of variance corresponds to the variability of the data from the mean, and this value appears to be very small in the sample path shown.
The reason is the same as above, namely that by generating many paths each one will have its own characteristics and, as far as the selected path is concerned, it appears to be less variable
- **Standard Deviation:** This value indicates the extent to which the forecasts deviate from the mean value, and in the case of the path under analysis we can see that these appear to be less distant from the mean value.
This seems completely understandable since, as mentioned, the standard deviation is the square root of the variance. Consequently, less variability in the data will be reflected in less distance between the estimated values and the mean value of the time series
- **Kurtosis:** As can be seen, the kurtosis value of the predicted time series appears to be higher than that of the historical data, indicating a higher concentration of extreme values.
This finds its explanation in the fact that the original time series had residuals with non-normal distribution, and this characteristic was amplified in the simulation of the path under analysis

- Skewness: as can be seen from the graph and, consistent with what was seen during the Kurtosis comparison, the extent of skewness between the original and predicted time series appears to be slightly greater in the latter, clearly showing a positive asymmetry corresponding to elongation of the right tail of the distribution and thus the presence of a greater number of extreme values.

Finally, the last step in this comparison was to compare the estimates obtained through ARMA (1,1) and those obtained through the Monte Carlo simulation represented by the median path.

Mean	Variance	Standard Deviation	Kurtosis	Skewness
196.33598958441027	144169.34060902166	379.69637950475857	4.873604843026309	2.3693826560711093

Figure 39: ARMA main statistics of the forecasted wind energy production

Mean	Variance	Standard Deviation	Kurtosis	Skewness
190.7237409971359	127672.8030836949	357.31331221169876	6.499735887207874	2.5976987422304703

Figure 40: Monte Carlo median path wind energy production: main statistical measures

The analysis of the above tables clearly indicates that:

- Mean: The value of this measure appears to be higher in the ARMA process indicating that this predicts higher wind power production of about 6kW per hour over the next 10 years
- Variance: The higher output of the ARMA model seen above is correlated with a higher variance which indicates that there will be greater variability in output than that predicted by the Monte Carlo simulation
- Standard Deviation: This measure is derived from the variance. As a result it shows how the values predicted by the ARMA model appear more dispersed than the average in comparison to the values predicted by the Monte Carlo simulation
- Kurtosis: this measure shows how "heavy" the tails of the distribution are and clearly shows how the values from the ARMA model appear to be more concentrated than the values predicted through Monte Carlo simulation, which, on the contrary, appear to be more present on the tails, which, in fact, appear to be heavier
- Skewness: as it was mentioned, this measure indicates the asymmetry of the distribution and, considering what was said during the kurtosis analysis, it shows and confirms that there is less elongation of the right tail in the time series obtained through the ARMA model than that obtained through the Monte Carlo simulation.

Chapter 6

Conclusion

The objective of this thesis, as stated at the beginning of the discussion, was to forecast wind production over a long-term time horizon in order to be able to include such projections in the process of defining a Power Purchase Agreement. Such contract, which are widespread in the market, appear to be of fundamental importance for both speculative and hedging purposes and play a key role in the development and deployment of energy from renewable sources since they make the cost programmable and controllable.

For the above contract to be used more frequently, and consequently green energy can experience a new phase of development, it is necessary to develop models able to predict the random features of the natural components that determine energy production. Such predictions have fundamental relevance not only for financial purposes (such as that of the creation of the aforementioned contract), but also for supply management and especially for what are called "peak loads".

In this thesis, in the purpose of predicting production, it was necessary to model wind speed, and two different approaches were used for this purpose, namely the ARMA process and Monte Carlo simulation. Both methodologies helped to explain the phenomenon and predict it using different underlying logic, equal to:

- The Monte Carlo simulation is based on the generation of a large number of simulations (based on random variables that follow a specific probabilistic distribution) randomized to allow exploration of possible configurations
- The ARMA process assumes the existence of a relationship between the observations in the time series and models this feature through its two components AR (Autoregressive) and MA (Moving Average)

With the forecasts calculated by these models, it is possible, by incorporating a dynamic of energy prices predicted for the same time horizon and with the same frequency, to define the design of a PPA, and thus to be able to negotiate this type of contract. In conclusion, a clarification is necessary.

Both models provided wind speed predictions that, combined with the use of the specific power curve of the plant, could be converted into wind production, and both models led to similar, but not equal predictions.

It is precisely this characteristic that is relevant. Indeed, if the models had led to identical results and if repeating the experiment n times had obtained for each time instant in the time series the same value it would have been logical to expect the predictions to be almost certain, but this is not the case, and this is precisely due to the error in the predictions, which, increases as the time horizon that is being

attempted to predict increases.

Precisely for this reason, this thesis (and especially its models, which are already dynamically structured through the use of APIs that can be easily modified, interpreted, and implemented) is intended to be a starting point for the development of further models and the application of further advanced mathematical-statistical-informatics techniques for forecasting the future.

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